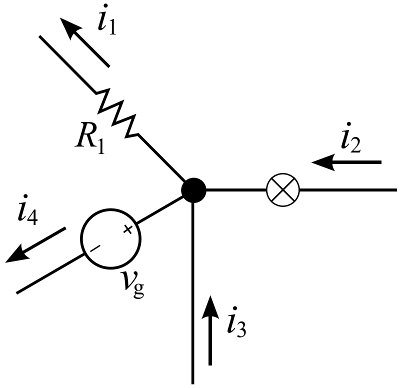
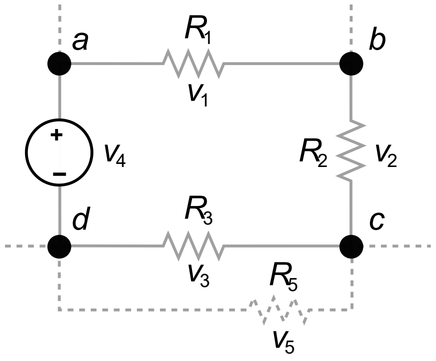
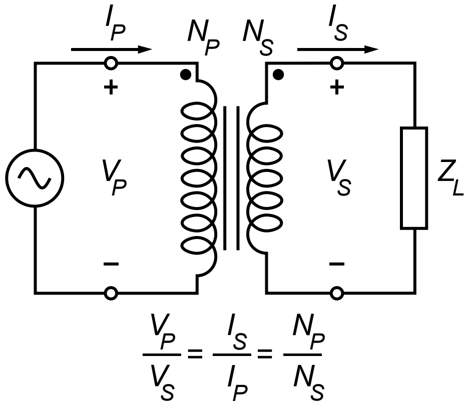
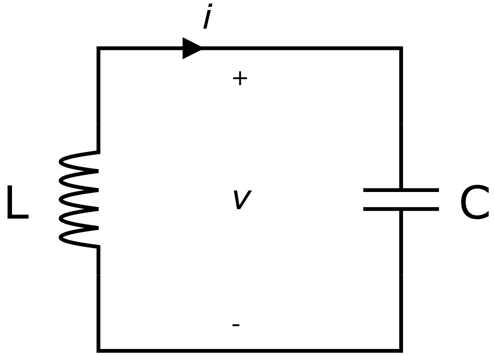
Introduction to Electrical Engineering









Overview

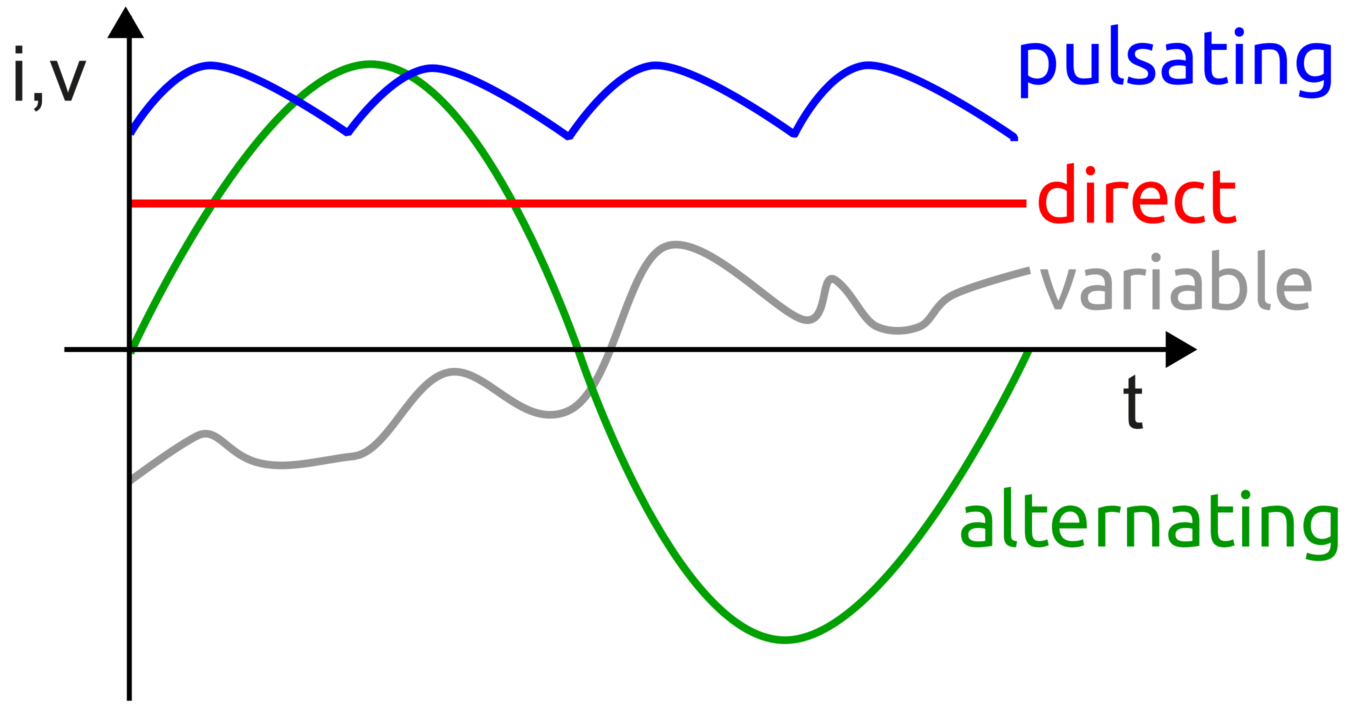
1. Physical Principles and Basic Concepts
2. Direct Current Circuits
3. Electric and Magnetic Fields
4. **Alternating Current Technology**
5. Wrap-Up and Outlook

Alternating Current Technology

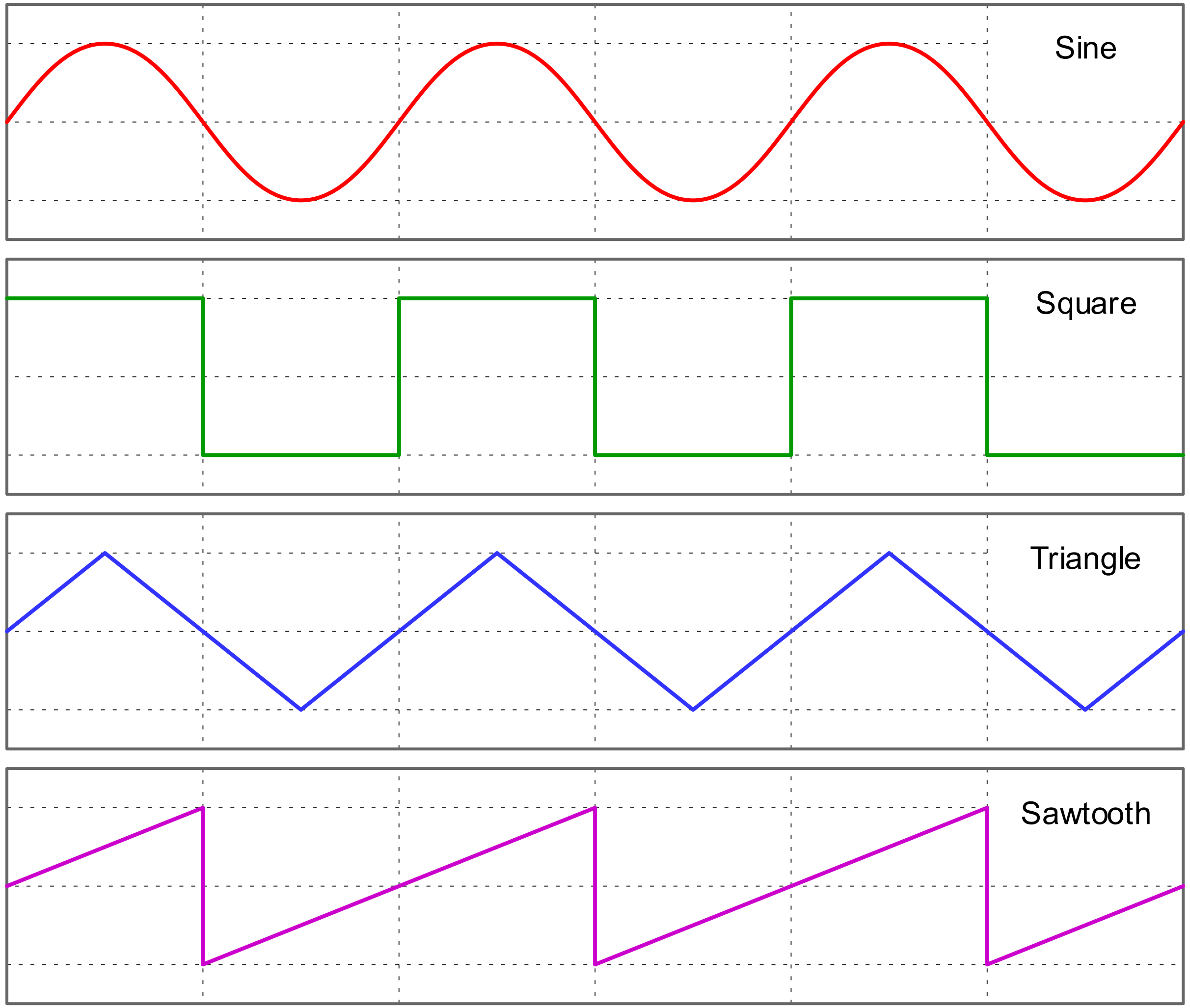
- Sinusoidal Waveform
- Phasor Representation
- Network Analysis in Phasor Domain
- AC Power
- High- & Low-Pass Filters
- Resonant Circuits
- Three-Phase Circuits

Alternating Current (AC)

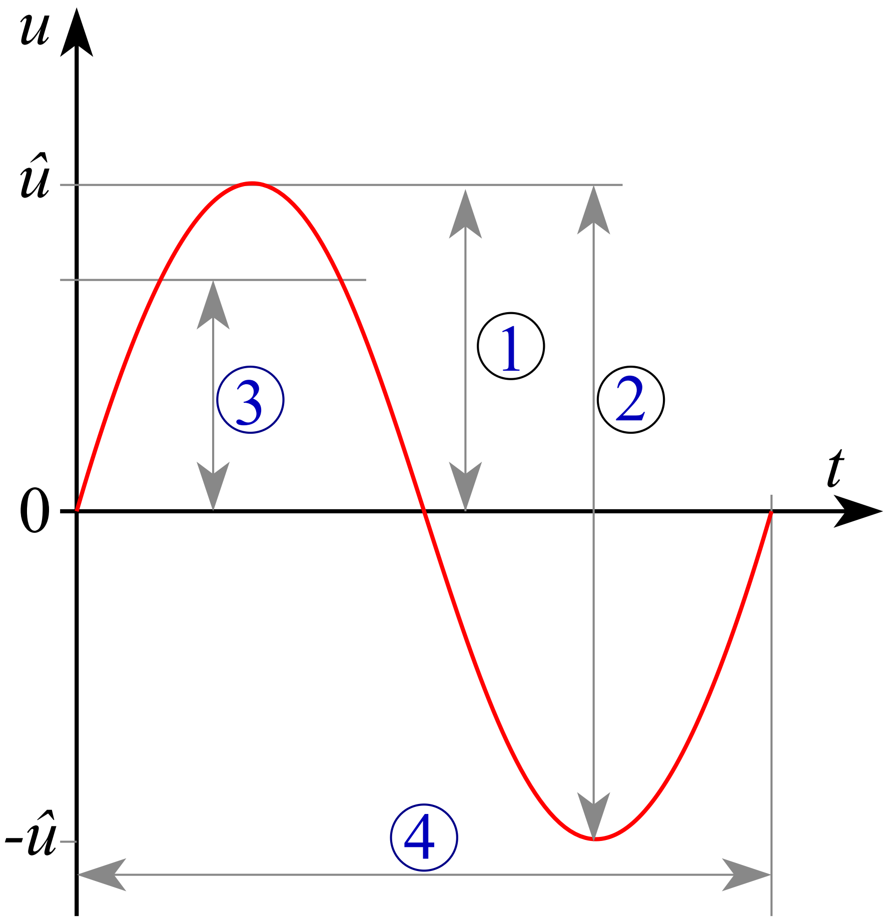
periodically reverses direction and changes its magnitude continuously with time



AC Waveforms



Sinusoidal AC



usual form of AC in most electric power circuits: sine wave

$u(t) = \hat{u}\, sin(\omega t), \quad i(t) = \hat{\imath}\, sin(\omega t)$

- $u(t) = u(t + nT)$
- frequency: $f = \frac{1}{T} \quad [f] = \frac{1}{\text{s}} = 1\ \text{Hz}$
- angular frequency: $\omega = 2\pi f \quad [\omega] = \frac{1}{s}$

1. peak voltage/current (amplitude) $\hat{u}, \hat{\imath}$
2. peak-to-peak amplitude: $2 \cdot \hat{u}$
3. RMS U, I
4. period (cycle time) T

Root Mean Square (RMS)

$$I = \sqrt{\overline{i^2}} = \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} i^2(t)dt} = \frac{\hat{\imath}}{\sqrt{2}}$$

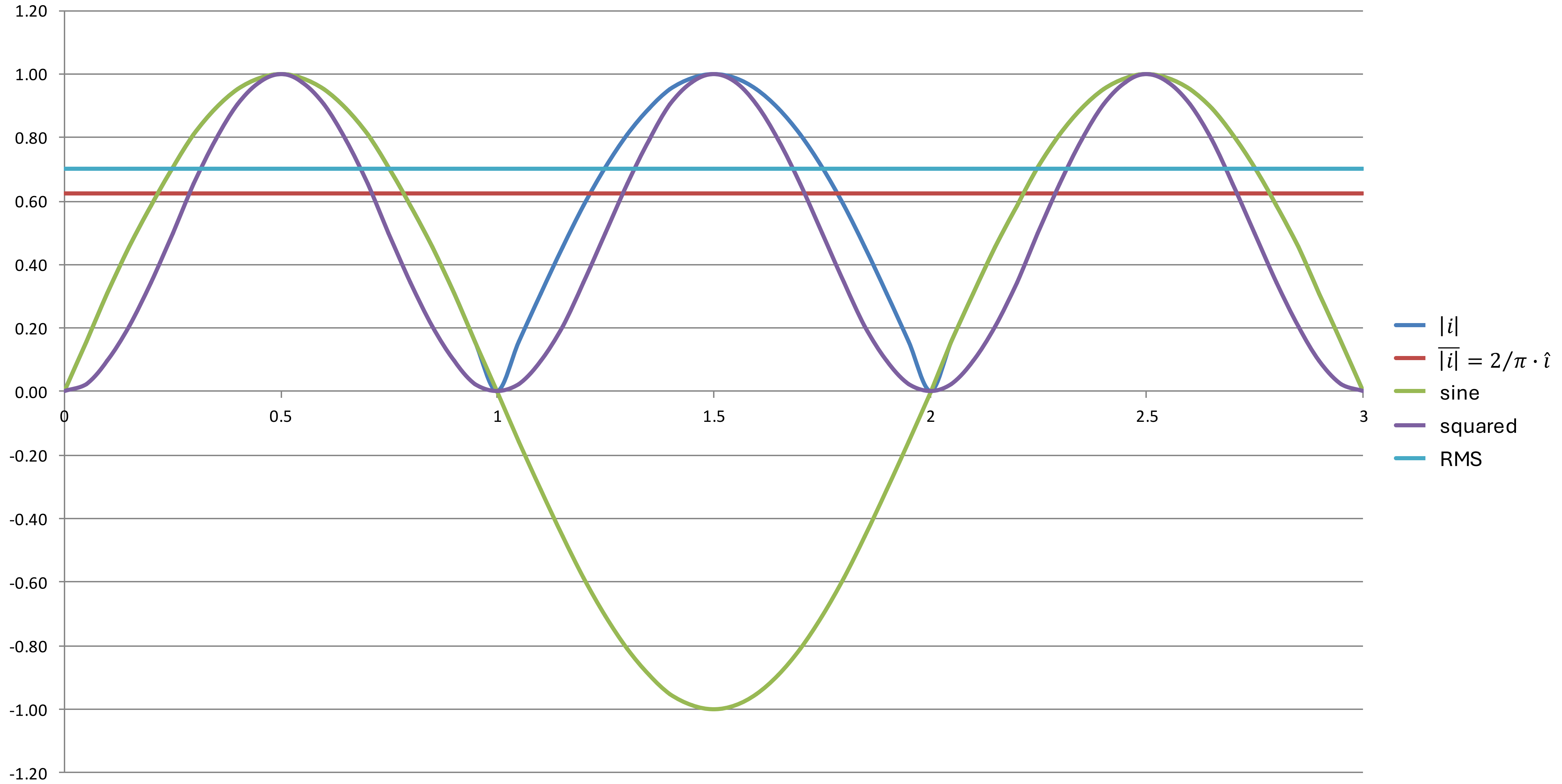
RMS of sum of mixed-frequency currents:

$$I_{\text{sum}} = \sqrt{I^2(f_1) + I^2(f_2)} \neq I(f_1) + I(f_2)$$

The RMS of an AC corresponds to the DC value that generates the same amount of heat in a resistor. → effective value

mean itself is zero for pure AC:

$$\bar{\imath} = \frac{1}{T}\int_{t_0}^{t_0+T} i(t)dt = 0$$


1.20
1.00
0.80
0.60
0.40
0.20
0.00
-0.20
-0.40
-0.60
-0.80
-1.00
-1.20
0
0.5
1
1.5
2
2.5
3
$|i|$
$\overline{|i|} = 2/\pi \cdot \hat{\imath}$
sine
squared
RMS

Phase Shift

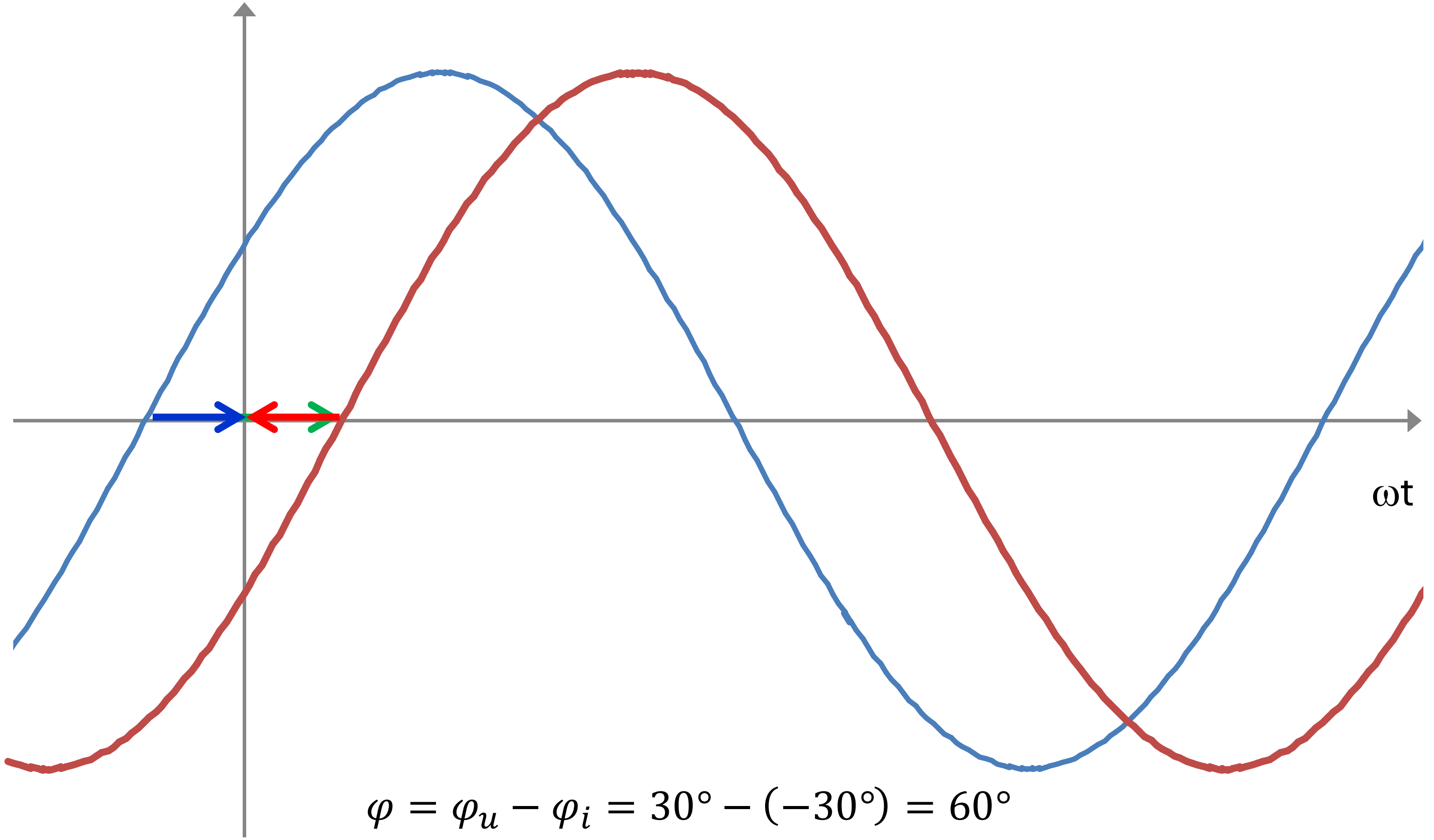
$$i(t) = \hat{\imath}\, sin(\omega t + \varphi_0)$$

phase shift (for two currents with same frequency):

$$\varphi = \varphi_{01} - \varphi_{02}$$

phase shift between voltage and current (with current as reference):

$$\varphi = \varphi_u - \varphi_i$$

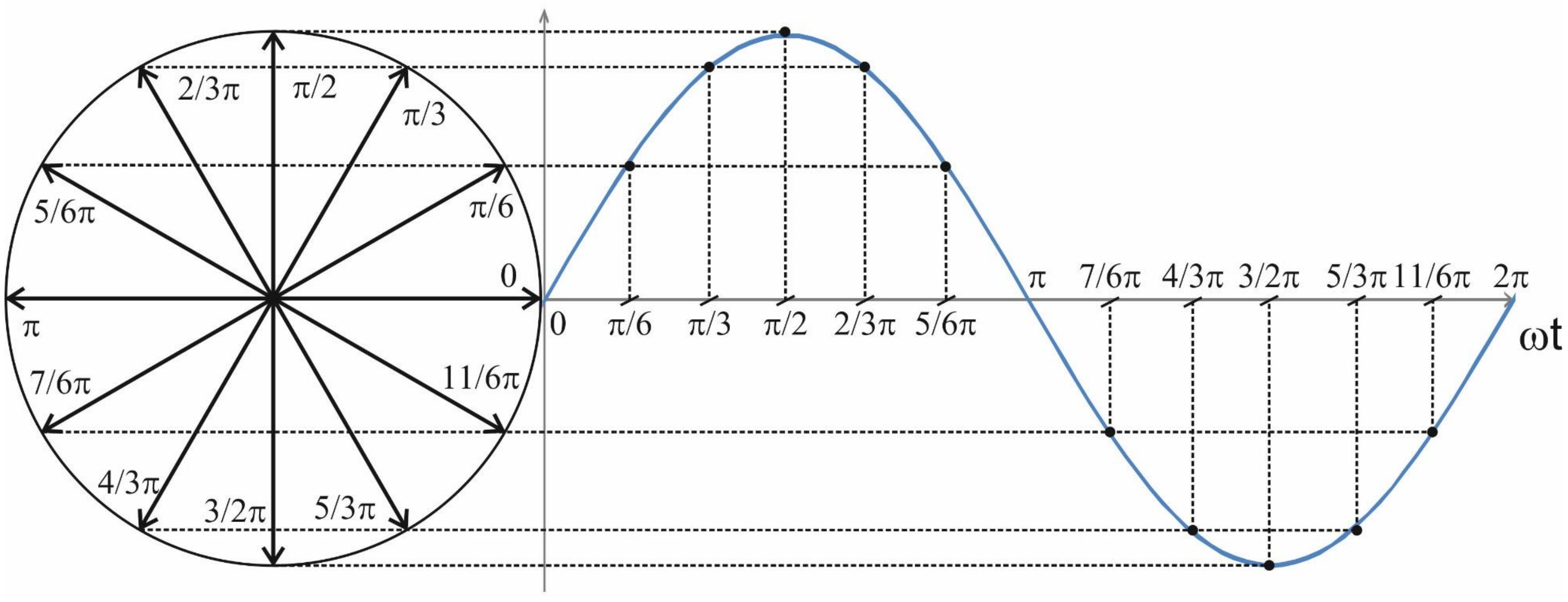


$$\varphi = \varphi_u - \varphi_i = 30° - (-30°) = 60°$$

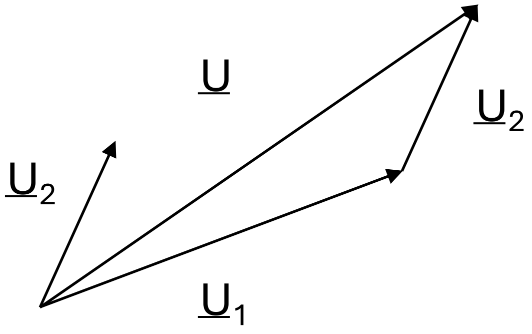
In this example, the current lags behind the voltage.

Phasor Representation

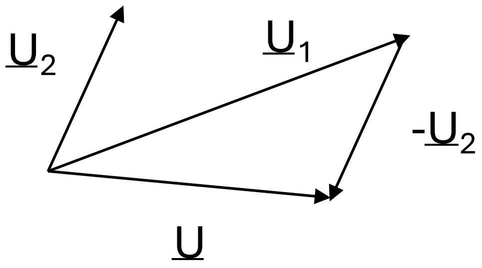
phasor diagram:



addition:



subtraction:



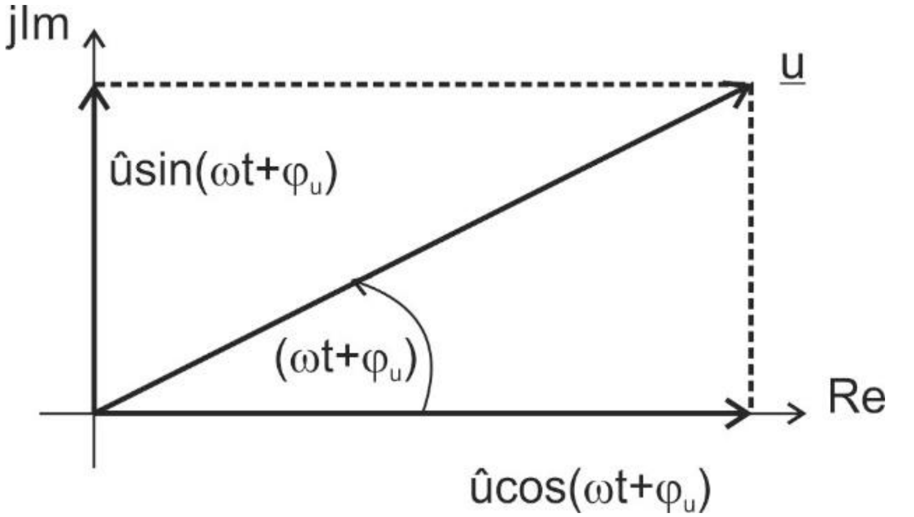
phasor calculations only work for quantities with same frequency
→ freeze rotation in time to get stationary phasors

Use of Complex Numbers

$$u(t) = \hat{u}\cos(\omega t + \varphi_u) = \hat{u}\sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right) = U \cdot \sqrt{2}\sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right)$$

phasors can be represented by complex numbers:

$$\underline{u}(t) = Re\left\{\underline{u}(t)\right\} + j\, Im\left\{\underline{u}(t)\right\} = \hat{u}\cos(\omega t + \varphi_u) + j\,\hat{u}\sin(\omega t + \varphi_u)$$



Complex Amplitude

$$\underline{u}(t) = \hat{u}[\cos(\omega t + \varphi_u) + j\sin(\omega t + \varphi_u)] = \hat{u}\ e^{j(\omega t + \varphi_u)} = \hat{u}\ e^{j\varphi_u}\underbrace{e^{j\omega t}}$$

complex

can be ignored if all quantities have the same frequency
→ stationary phasors

complex amplitude: $\underline{\hat{u}} = \hat{u}\ e^{j\varphi_u}$

$$\underline{u}(t) = \hat{u}\ e^{j\varphi_u}e^{j\omega t} = \underline{\hat{u}}\ e^{j\omega t}$$

complex RMS: $\underline{U} = U\ e^{j\varphi_u} = \frac{\underline{\hat{u}}}{\sqrt{2}}$

Transformation into Phasor Domain

time-dependent voltage	complex amplitude
$\hat{u}\cos(\omega t)$	$\underline{\hat{u}} = \hat{u}$
$\hat{u}\cos(\omega t + \varphi_u)$	$\underline{\hat{u}} = \hat{u}\, e^{j\varphi_u}$
$\hat{u}\sin(\omega t) = \hat{u}\cos\left(\omega t - \frac{\pi}{2}\right)$	$\underline{\hat{u}} = \hat{u}\, e^{-j\frac{\pi}{2}}$
$\hat{u}\sin(\omega t + \varphi_u) = \hat{u}\cos\left(\omega t + \varphi_u - \frac{\pi}{2}\right)$	$\underline{\hat{u}} = \hat{u}\, e^{j\left(\varphi_u - \frac{\pi}{2}\right)}$

$$\underline{\hat{u}} = \hat{u}(\cos\varphi_u + j\sin\varphi_u) = Re + jIm$$

Back Transformation into Time Domain

1. multiplication with $e^{j\omega t}$ (phasors are stationary)

2. consider only real part of the complex number (or imaginary part if sine was used instead of cosine)

Derivation in Phasor Domain

$$\frac{d}{dt} \rightarrow j\omega \qquad \left(\frac{d}{dt}e^{j\omega t} = e^{j\omega t} \cdot j\omega\right)$$

→ phase shift of $\boldsymbol{\frac{\pi}{2}}$ $\quad \left(\underline{\hat{u}} \cdot j = \hat{u}\ e^{j\varphi_u} \cdot e^{j\boldsymbol{\frac{\pi}{2}}} = \hat{u}\ e^{j\left(\varphi_u + \boldsymbol{\frac{\pi}{2}}\right)}\right)$

example:

$$u(t) = L\frac{di}{dt} \rightarrow \underline{U} = j\omega L\underline{I}$$

Integration in Phasor Domain

$$\int dt \rightarrow \frac{1}{j\omega}$$

$$\left(\int e^{j\omega t}\, dt = e^{j\omega t} \cdot \frac{1}{j\omega}\right)$$

→ phase shift of $-\frac{\boldsymbol{\pi}}{\mathbf{2}}$

$$\left(\underline{\hat{u}}/j = \hat{u}\; e^{j\varphi_u} \cdot e^{-j\frac{\boldsymbol{\pi}}{\mathbf{2}}} = \hat{u}\; e^{j\left(\varphi_u - \frac{\boldsymbol{\pi}}{\mathbf{2}}\right)}\right)$$

example:

$$i(t) = \frac{1}{L}\int u(t)dt \rightarrow \underline{I} = \frac{1}{j\omega L}\underline{U}$$

Complex Resistance

impedance:

$$\frac{\underline{u}(t)}{\underline{i}(t)} = \frac{\underline{\hat{u}}\, e^{j\omega t}}{\underline{\hat{\imath}}\, e^{j\omega t}} = \frac{|\underline{\hat{u}}| e^{j\varphi_u}}{|\underline{\hat{\imath}}| e^{j\varphi_i}} = |\underline{Z}| e^{j(\varphi_u - \varphi_i)} = |\underline{Z}| e^{j\varphi} = \underline{Z}$$

abstract quantity

Ohm‘s law in phasor domain: $\underline{\hat{u}} = \underline{Z}\, \underline{\hat{\imath}}$ or $\underline{U} = \underline{Z}\, \underline{I}$

$$\underline{Z} = R + jX = |\underline{Z}| e^{j\varphi}$$

impedance angle

resistance

reactance

modulus of impedance
(apparent impedance)

$$|\underline{Z}| = \sqrt{R^2 + X^2}$$

$$\tan\varphi = \frac{X}{R}$$

Complex Conductance

admittance:

abstract quantity

$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{R + jX} = \underbrace{\frac{R}{R^2 + X^2}}_{\text{conductance } G} + j\underbrace{\frac{-X}{R^2 + X^2}}_{\text{susceptance } B}$$

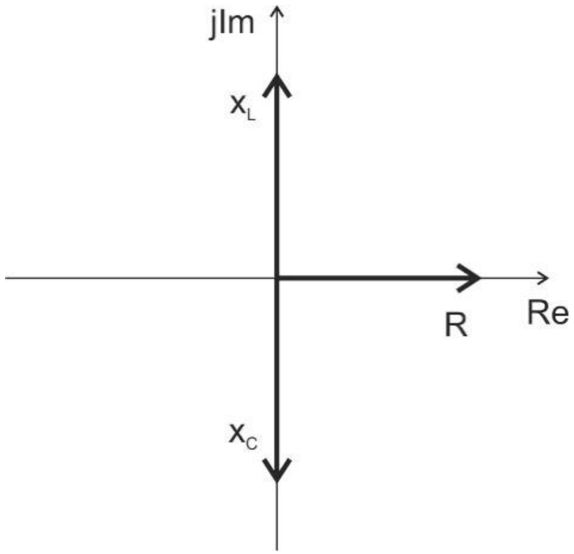
$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{|\underline{Z}|e^{j\varphi}} = |\underline{Y}|e^{-j\varphi}$$

admittance angle
$\theta = -\varphi$

modulus of admittance
(apparent admittance)

Linear Passive Components

R	$\underline{\hat{u}} = R \cdot \underline{\hat{\imath}}$	$\underline{Z} = R$
L	$\underline{\hat{u}} = j\omega L \cdot \underline{\hat{\imath}}$	$\underline{Z}_L = j\omega L = jX_L$
C	$\underline{\hat{u}} = \frac{1}{j\omega C} \cdot \underline{\hat{\imath}}$	$\underline{Z}_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = jX_C$

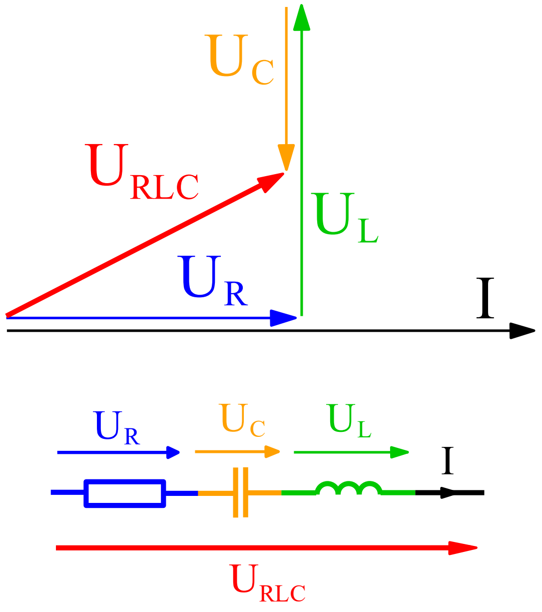


R: U and I in phase ($\varphi = 0$)

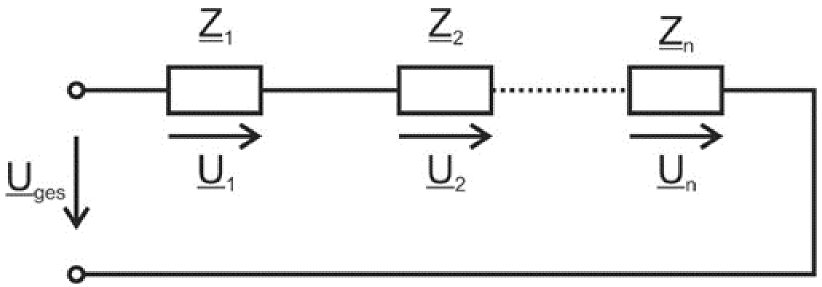
L: U leads I by 90°

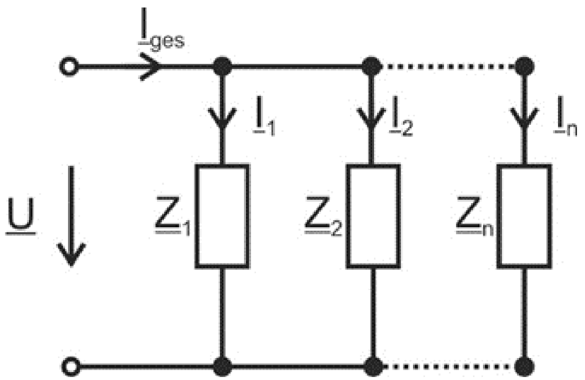
C: U lags I by 90°

must consider angles:



Network Analysis in Phasor Domain





$$\underline{Z} = \sum_{k=1}^{n} \underline{Z}_k$$

$$\frac{1}{\underline{Z}} = \sum_{k=1}^{n} \frac{1}{\underline{Z}_k}$$

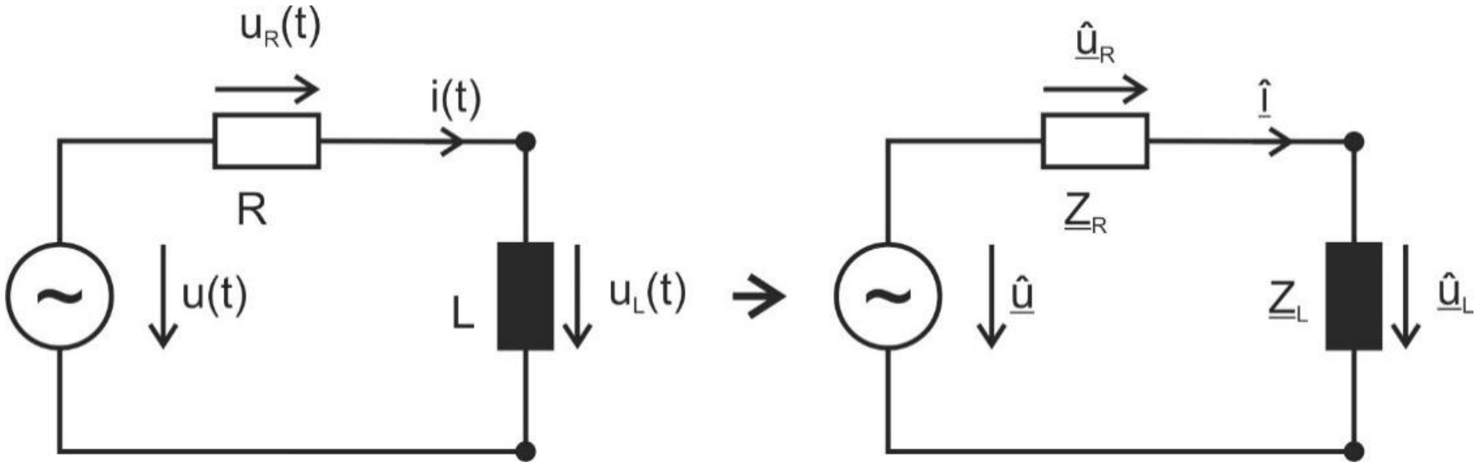
$$\frac{\underline{U}_1}{\underline{U}_2} = \frac{\underline{Z}_1}{\underline{Z}_2}$$

$$\frac{\underline{I}_1}{\underline{I}_2} = \frac{\underline{Z}_2}{\underline{Z}_1}$$

KCL and KVL also valid for complex amplitudes and RMS values

Example 1

The current through the circuit and the voltage across the coil are to be determined.



$$u(t) = \hat{u}\cos(\omega t + \varphi_u) \quad \rightarrow \quad \underline{\hat{u}} = \hat{u}e^{j\varphi_u}$$

Example 1

$$\underline{\hat{u}} = \underline{\hat{u}}_R + \underline{\hat{u}}_L = \underline{Z_R} \cdot \underline{\hat{\imath}} + \underline{Z_L} \cdot \underline{\hat{\imath}} = (R + j\omega L)\underline{\hat{\imath}}$$

$$\underline{\hat{\imath}} = \frac{\underline{\hat{u}}}{R + j\omega L} = \frac{\underline{\hat{u}} \cdot (R - j\omega L)}{R^2 + \omega^2 L^2} = \underline{\hat{u}}\frac{R}{R^2 + \omega^2 L^2} - j\underline{\hat{u}}\frac{\omega L}{R^2 + \omega^2 L^2}$$

$$\hat{\imath} = \frac{\hat{u}}{Z} = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}}$$

$$\varphi = \tan^{-1}\frac{\omega L}{R} = \varphi_u - \varphi_i$$

$$\left(\varphi_u = \tan^{-1}\frac{\omega L}{R}; \quad \varphi_i = 0 \quad \text{or} \quad \varphi_u = 0; \quad \varphi_i = \tan^{-1}\frac{-\omega L}{R}\right)$$

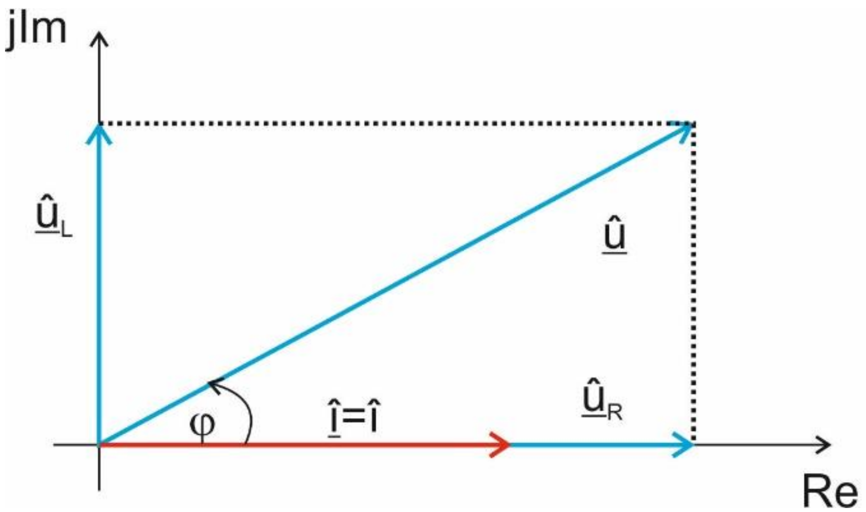
$$\underline{\hat{\imath}} = \hat{\imath} \cdot e^{j\varphi_i} = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}} e^{j(\varphi_u - \varphi)}$$

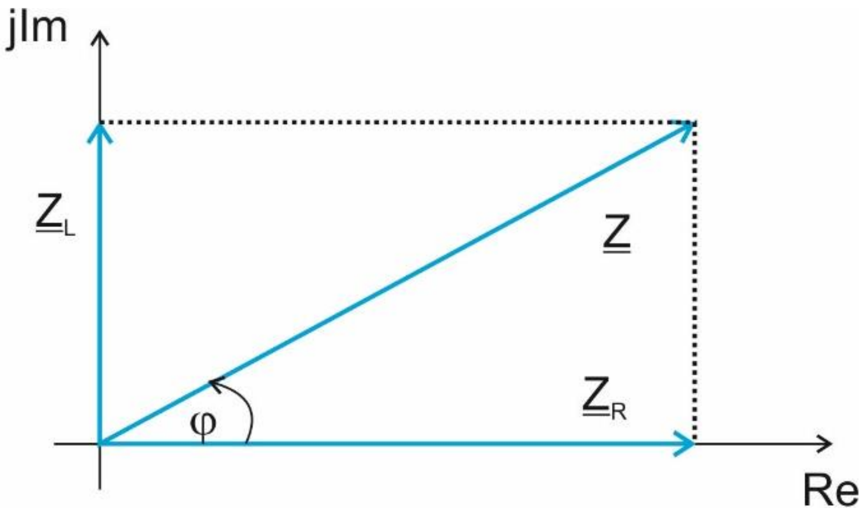
Example 1

$$i(t) = Re\{\underline{\hat{\imath}} \cdot e^{j\omega t}\} = Re\{\hat{\imath} \cdot e^{j\varphi_i} e^{j\omega t}\} = \hat{\imath}\cos(\omega t + \varphi_i) = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}}\cos\left(\omega t + \varphi_u - \tan^{-1}\frac{\omega L}{R}\right)$$

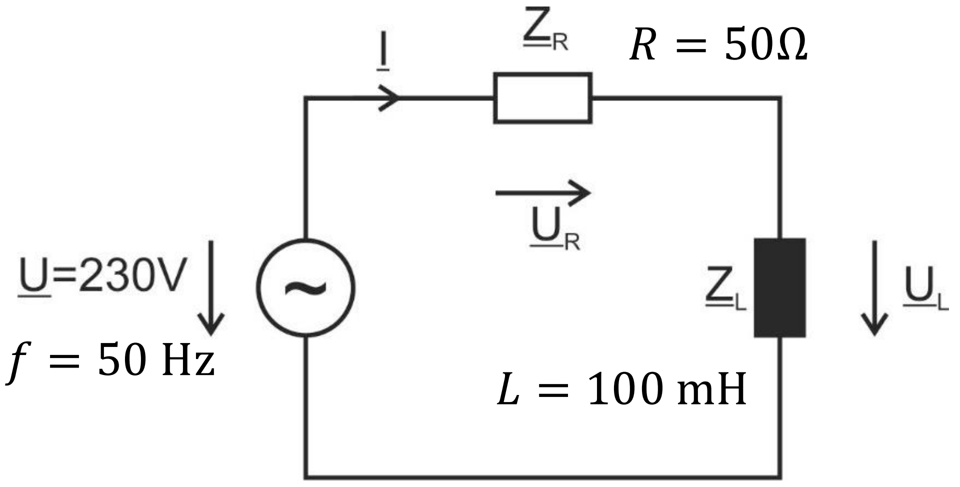
$$\underline{\hat{u}}_L = \underline{Z}_L \cdot \underline{\hat{\imath}} = j\omega L\underline{\hat{\imath}} = e^{j\frac{\pi}{2}}\omega L\hat{\imath}e^{j\varphi_i} = \hat{\imath}\omega L e^{j\left(\varphi_i + \frac{\pi}{2}\right)}$$

$$u_L(t) = Re\{\underline{\hat{u}}_L e^{j\omega t}\} = Re\left\{\hat{\imath}\omega L e^{j\left(\omega t + \varphi_i + \frac{\pi}{2}\right)}\right\} = \hat{u}\frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}\cos\left(\omega t + \frac{\pi}{2} + \varphi_u - \tan^{-1}\frac{\omega L}{R}\right)$$





Example 2



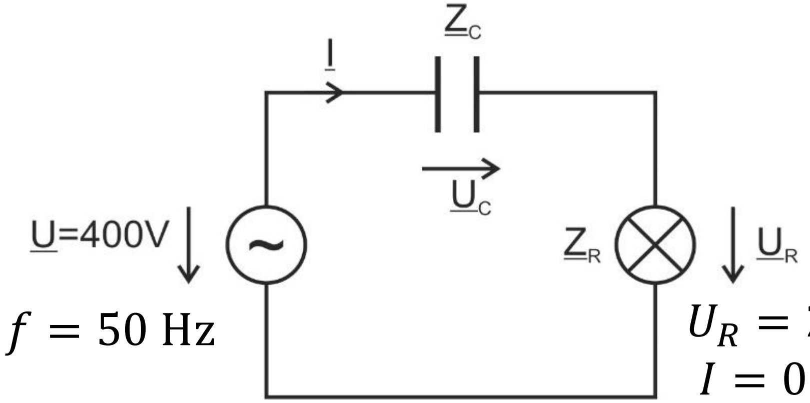
$$\underline{Z} = \underline{Z_R} + \underline{Z_L} = R + j\omega L = (50 + j\ 31.42)\ \Omega$$
$$Z = \sqrt{R^2 + (\omega L)^2} = 59.1\ \Omega$$
$$\varphi = \tan^{-1}\frac{\omega L}{R} = 32.1°$$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = \frac{U}{(R + j\omega L)} = \frac{U(R - j\omega L)}{(R^2 + (\omega L)^2)} = (3.30 - j\ 2.07)\mathrm{A} = 3.90\ \mathrm{A} \cdot e^{-j\ 32.1°}$$

$$\underline{U_R} = \underline{Z_R} \cdot \underline{I} = 50\ \Omega \cdot 3.90\ \mathrm{A} \cdot e^{-j\ 32.1°} = 195\ \mathrm{V} \cdot e^{-j\ 32.1°} = (165.19 - j\ 103.6)\ \mathrm{V}$$

$$\underline{U_L} = \underline{Z_L} \cdot \underline{I} = j\omega L \cdot \underline{I} = j\ 31.42\ \Omega \cdot (3.30 - j\ 2.07)\ \mathrm{A} = (65.04 + j\ 103.69)\ \mathrm{V} = 122.38\ \mathrm{V} \cdot e^{j\ 57.9°}$$

Example 3



$f = 50$ Hz

$U_R = 230$ V
$I = 0.27$ A

What capacitance must the capacitor have so that the light bulb is connected to the voltage $U_R = 230$ V?

$$U^2 = {U_C}^2 + {U_R}^2 \;\rightarrow\; U_C = \sqrt{400^2 - 230^2}\ \text{V} = 327.26\ \text{V}$$

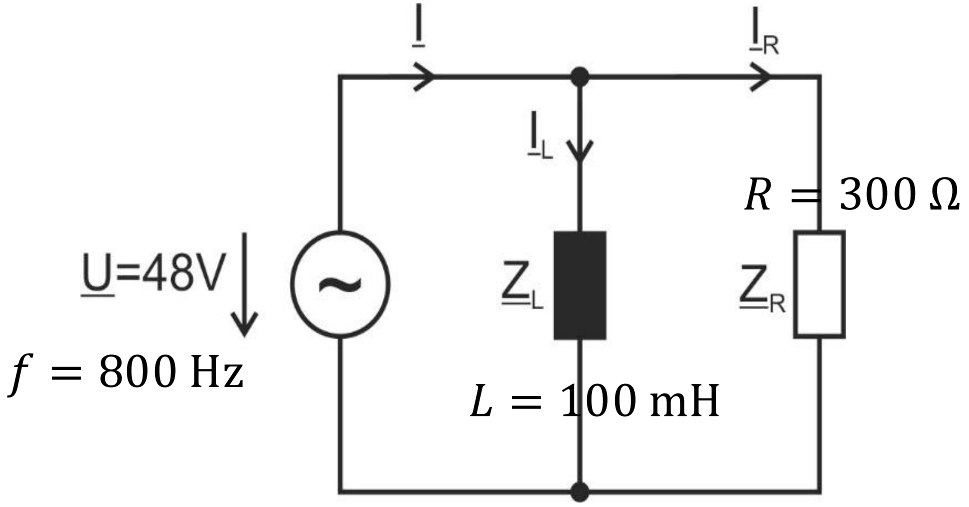
$$\frac{1}{\omega C} = \frac{U_C}{I} \;\rightarrow\; C = 2.63\ \mu\text{F}$$

Example 4

$$\frac{1}{\underline{Z}} = \frac{1}{R} + \frac{1}{j\omega L} = \frac{j\omega L + R}{j\omega RL}$$

$$\rightarrow \underline{Z} = \frac{\omega^2 RL^2 + j\omega R^2 L}{R^2 + (\omega L)^2} = (221.2 + j\ 132.0)\ \Omega$$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = (160 - j\ 95)\ \mathrm{mA}$$



Instantaneous Power

$$p(t) = u(t) \cdot i(t)$$

For a purely ohmic resistance, current and voltage are in phase:

$$p(t) = \hat{u} \sin \omega t \cdot \hat{\imath} \sin \omega t = \hat{u}\hat{\imath}\; sin^2 \omega t$$
$$= \hat{u}\hat{\imath}\frac{1}{2}(1 - \cos 2\omega t) = UI(1 - \cos 2\omega t)$$

$p(t) > 0$ for passive sign convention → passive component

Average Power

area under power curve: electrical energy converted into heat in a resistor

average power (aka active power for resistance):

$$\bar{P} = \frac{1}{T}\int_0^T p(t)dt$$

resistance

$$= \frac{1}{T}\int_0^T \frac{1}{2}\hat{u}\hat{\imath}(1 - \cos 2\omega t)dt = \frac{1}{2}\hat{u}\hat{\imath}\frac{1}{T}\left(t - \frac{1}{2\omega}\sin 2\omega t\right)\Bigg|_0^T$$

$$= \frac{1}{2}\hat{u}\hat{\imath}\frac{1}{T}\left(T - \frac{1}{2\omega}\sin 2\omega T\right) = \frac{1}{2}\hat{u}\hat{\imath}\frac{1}{T}\left(T - \frac{1}{2\omega}\sin 2\omega\frac{2\pi}{\omega}\right) = \frac{1}{2}\hat{u}\hat{\imath} = UI$$

Reactive Power

for reactance (capacitor or inductor): phase shift of $\frac{\pi}{2}$ between $u(t)$ and $i(t)$

$$p(t) = \hat{u}\sin\omega t \cdot \hat{\imath}\sin\left(\omega t + \frac{\pi}{2}\right) = \hat{u}\hat{\imath}\sin\omega t\cos\omega t = \hat{u}\hat{\imath}\frac{1}{2}\sin 2\omega t = UI\sin 2\omega t$$

$$\bar{P} = \frac{1}{T}\int_0^T p(t)dt = \frac{1}{T}\int_0^T \frac{1}{2}\hat{u}\hat{\imath}(\sin 2\omega t)dt = \frac{1}{2}\hat{u}\hat{\imath}\frac{1}{T}\left(-\frac{1}{2\omega}\cos 2\omega t\right)\Bigg|_0^T$$

$$= \frac{1}{2}\hat{u}\hat{\imath}\frac{1}{T}\left(-\frac{1}{2\omega}(\cos 2\omega T + 1)\right) = \frac{1}{2}\hat{u}\hat{\imath}\frac{1}{T}\left(-\frac{1}{2\omega}(\cos 4\pi + 1)\right) = 0$$

→ not converted into heat or useful work

Apparent Power

mixed case for resistive-capacitive or resistive-inductive load:

$$p(t) = UI \cdot \cos\varphi \cdot (1 - \cos 2\omega t) - UI \cdot \sin\varphi \sin 2\omega t$$

- purely active for $\varphi = 0$
- purely reactive for $\varphi = \frac{\pi}{2}$ or $\varphi = -\frac{\pi}{2}$

average power: $P = UI \cdot \cos\varphi$

apparent power: $S = U \cdot I$

$$[S] = 1\,\mathrm{VA}$$

electrical devices must be designed for the apparent power to be transmitted

Complex Power

$$\underline{S} = \underline{U} \cdot \underline{I}^* = Ue^{j\varphi_u} Ie^{-j\varphi_i} = Se^{j\overbrace{(\varphi_u - \varphi_i)}^{\varphi}}$$

$$\underline{S} = \underline{U}\left(\frac{\underline{U}}{\underline{Z}}\right)^* = \underline{U} \cdot \underline{U}^* \frac{1}{\underline{Z}^*} = \frac{U^2}{\underline{Z}^*}$$

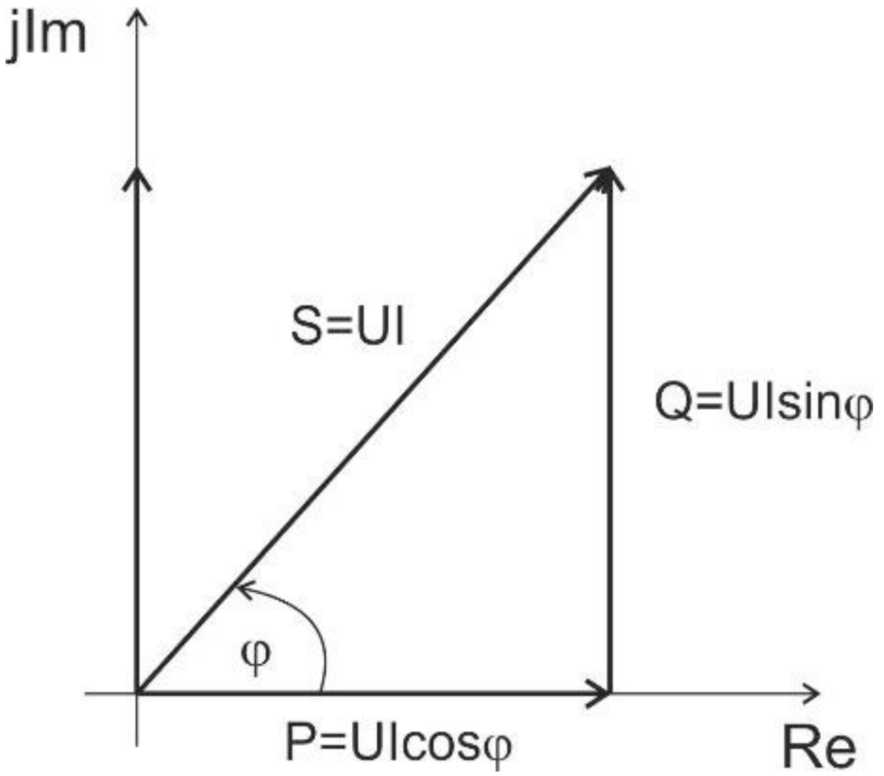
$$\underline{S} = \frac{1}{2}\underline{\hat{u}} \cdot \underline{\hat{\imath}}^* = P + jQ$$

active power

reactive power

with $S = \sqrt{P^2 + Q^2} = U \cdot I$

apparent power

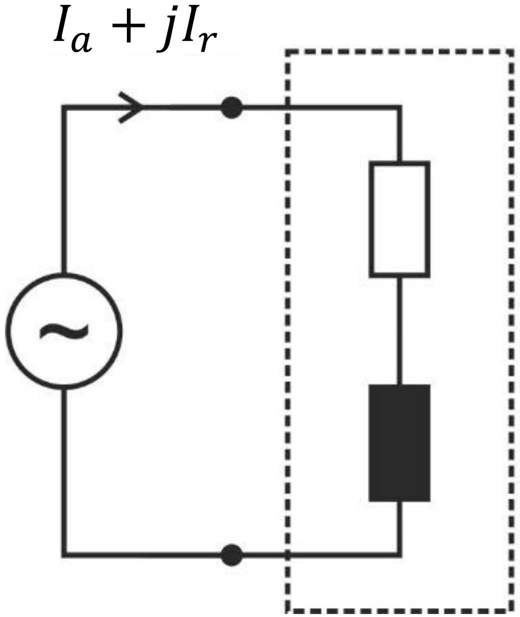


$$\varphi = \tan^{-1}\frac{Q}{P}; \quad \cos\varphi = \frac{P}{S}; \quad \sin\varphi = \frac{Q}{S}$$

Reactive Power Compensation

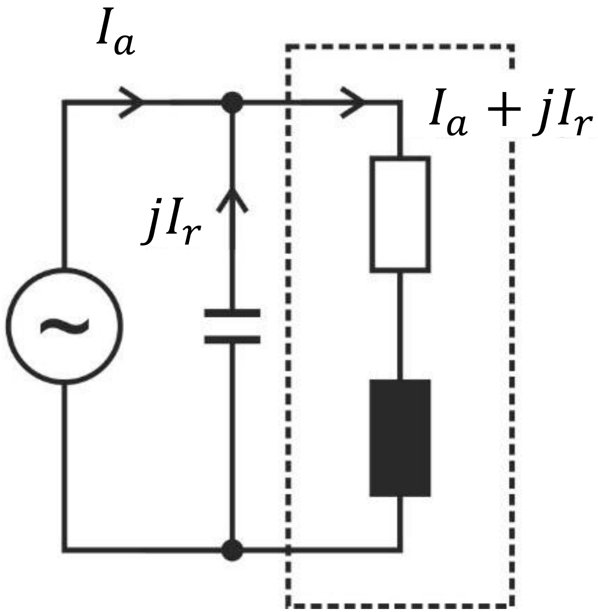
The absorption of inductive reactive power corresponds to the output of capacitive reactive power and vice versa: $\varphi = 180°$

→ can compensate each other (useful to avoid transmission losses)



resistive-inductive load

can be compensated by capacitor



Example

A motor ($U = 230\ \mathrm{V}$, $f = 50\ \mathrm{Hz}$, $I = 16\ \mathrm{A}$, $\cos\varphi = 0.8$) is to be compensated using a capacitor.

without capacitor:

$$I_a = I \cdot \cos\varphi = 12.8\ \mathrm{A},\ \ I_r = I \cdot \sin\varphi = -9.6\ \mathrm{A}\ \ \rightarrow \underline{I} = (12.8 - j\ 9.6)\ \mathrm{A}$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = 230\ \mathrm{V} \cdot (12.8 + j\ 9.6)\ \mathrm{A}$$

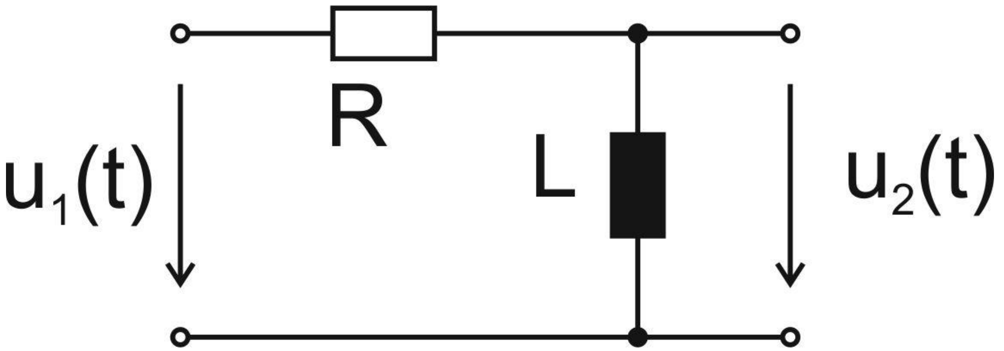
$$S = \sqrt{2.944^2 + 2.208^2}\ \mathrm{kVA} = 3.68\ \mathrm{kVA},\ \ \varphi = \tan^{-1}\frac{2.208}{2.944} = 36.87°$$

reactive current needs to be compensated by capacitor:

$$\frac{1}{\omega C} = \frac{U}{I_r}\ \ \rightarrow C = 133\ \mu\mathrm{F}$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = 230\ \mathrm{V} \cdot 12.8\ \mathrm{A} = 2.944\ \mathrm{kVA} = S,\ \ \varphi = 0$$

Frequency-Dependent Voltage Divider

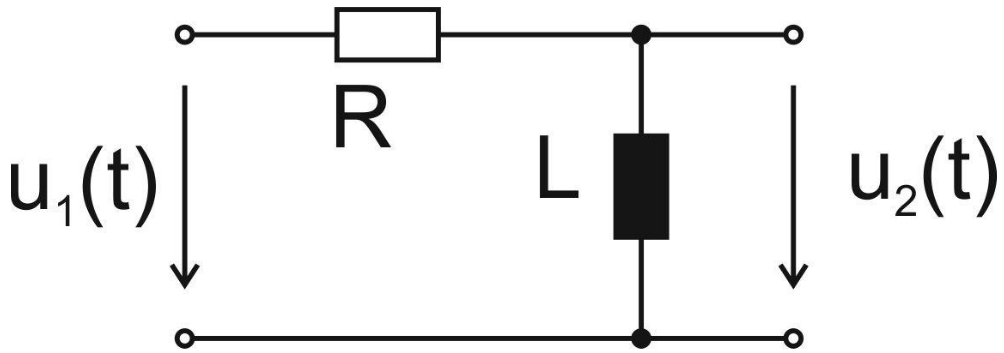


$$\underline{U}_2 = \underline{U}_1 \cdot \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \underline{U}_1 \cdot \frac{j\omega L}{R + j\omega L} = \underline{U}_1 \cdot \frac{(\omega L)^2 - j\omega LR}{R^2 + (\omega L)^2}$$

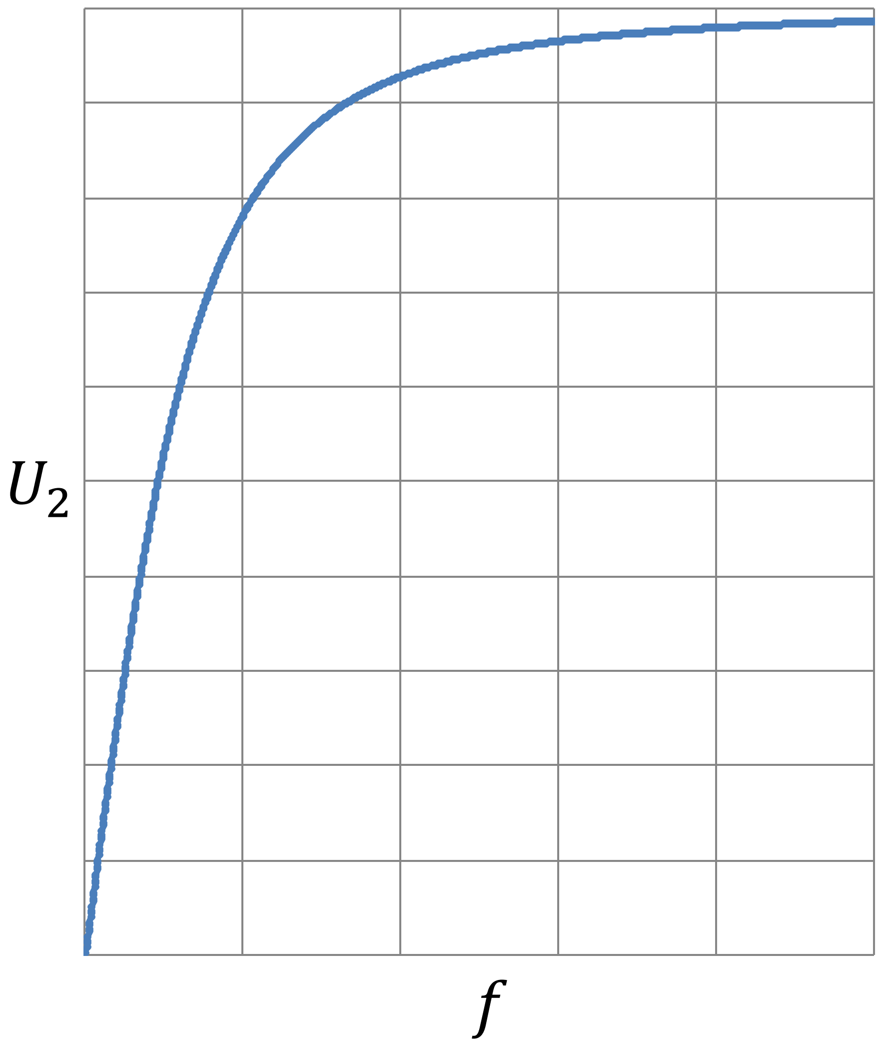
$$U_2 = U_1 \cdot \sqrt{\left(\frac{(\omega L)^2}{R^2 + (\omega L)^2}\right)^2 + \left(\frac{\omega LR}{R^2 + (\omega L)^2}\right)^2} = U_1 \cdot \sqrt{\frac{(\omega L)^4 + (\omega LR)^2}{(R^2 + (\omega L)^2)^2}}$$

$$= U_1 \frac{\sqrt{(\omega L)^2\left((\omega L)^2 + R^2\right)}}{R^2 + (\omega L)^2} = U_1 \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}} = U_1 \cdot \frac{Z_1}{Z}$$

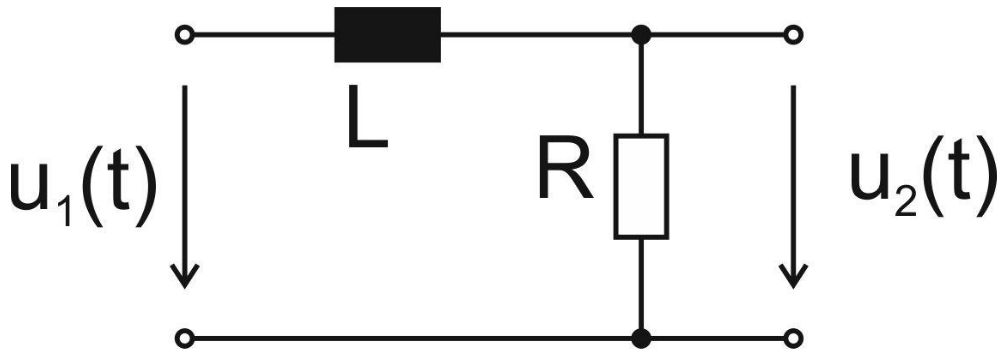
High-Pass RL Filter



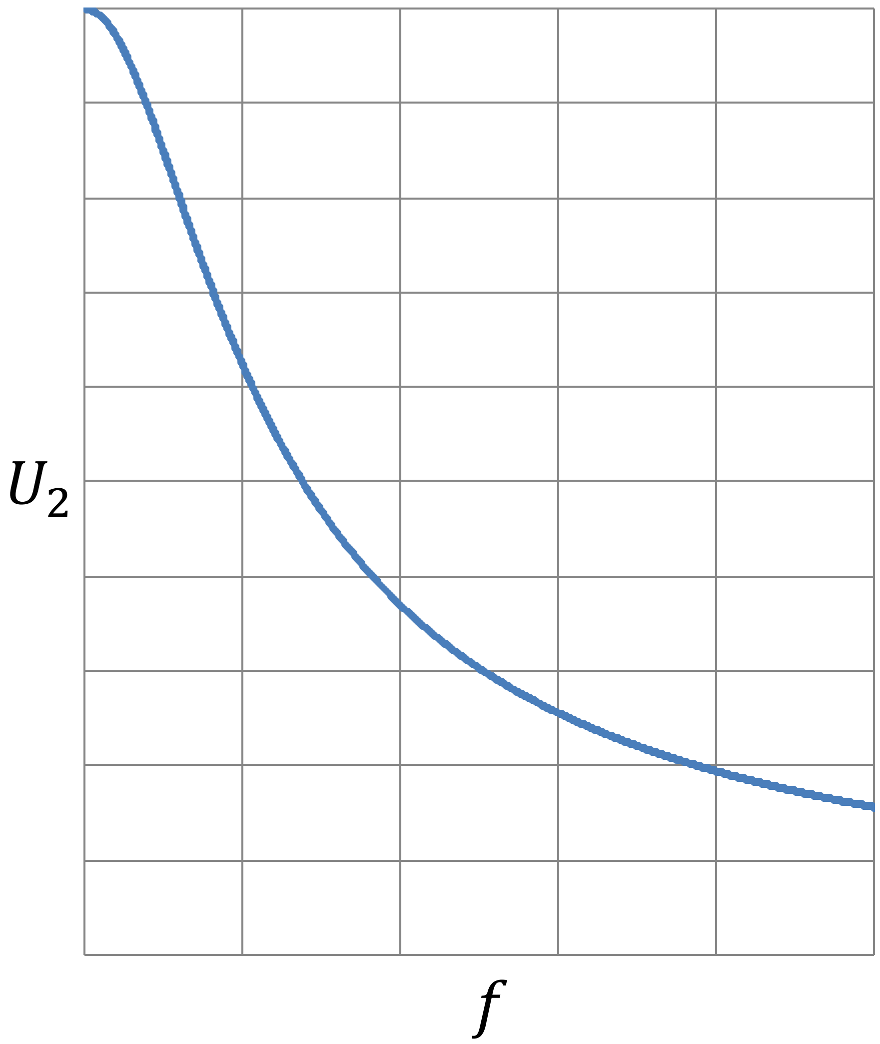
$$U_2 = U_1 \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}$$



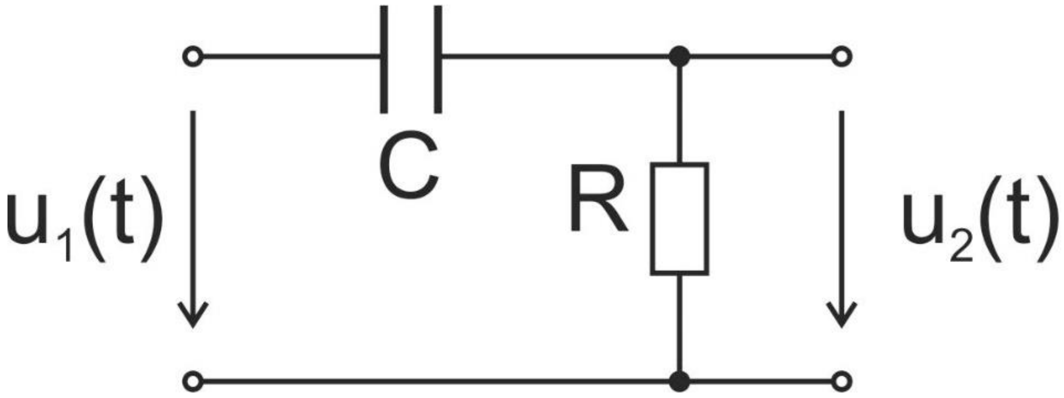
Low-Pass RL Filter



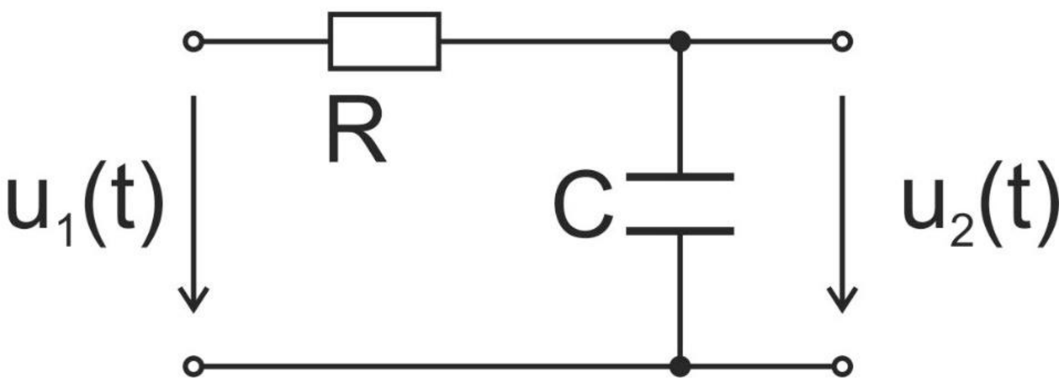
$$U_2 = U_1 \frac{R}{\sqrt{R^2 + (\omega L)^2}}$$



RC Filters



high-pass filter



low-pass filter

first-order filters: either a capacitor or an inductor

Series Resonant Circuit

$$\underline{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

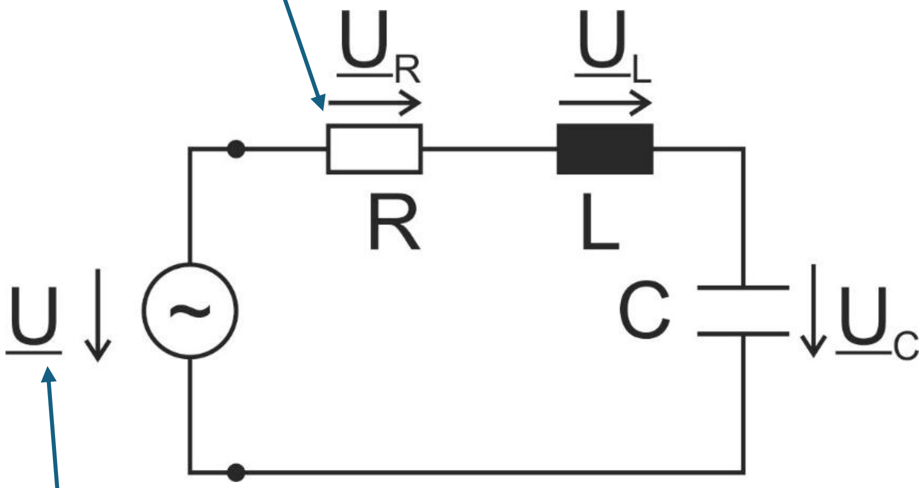
resonance frequency f_0: inductive and capacitive reactances cancel each other out

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\rightarrow \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

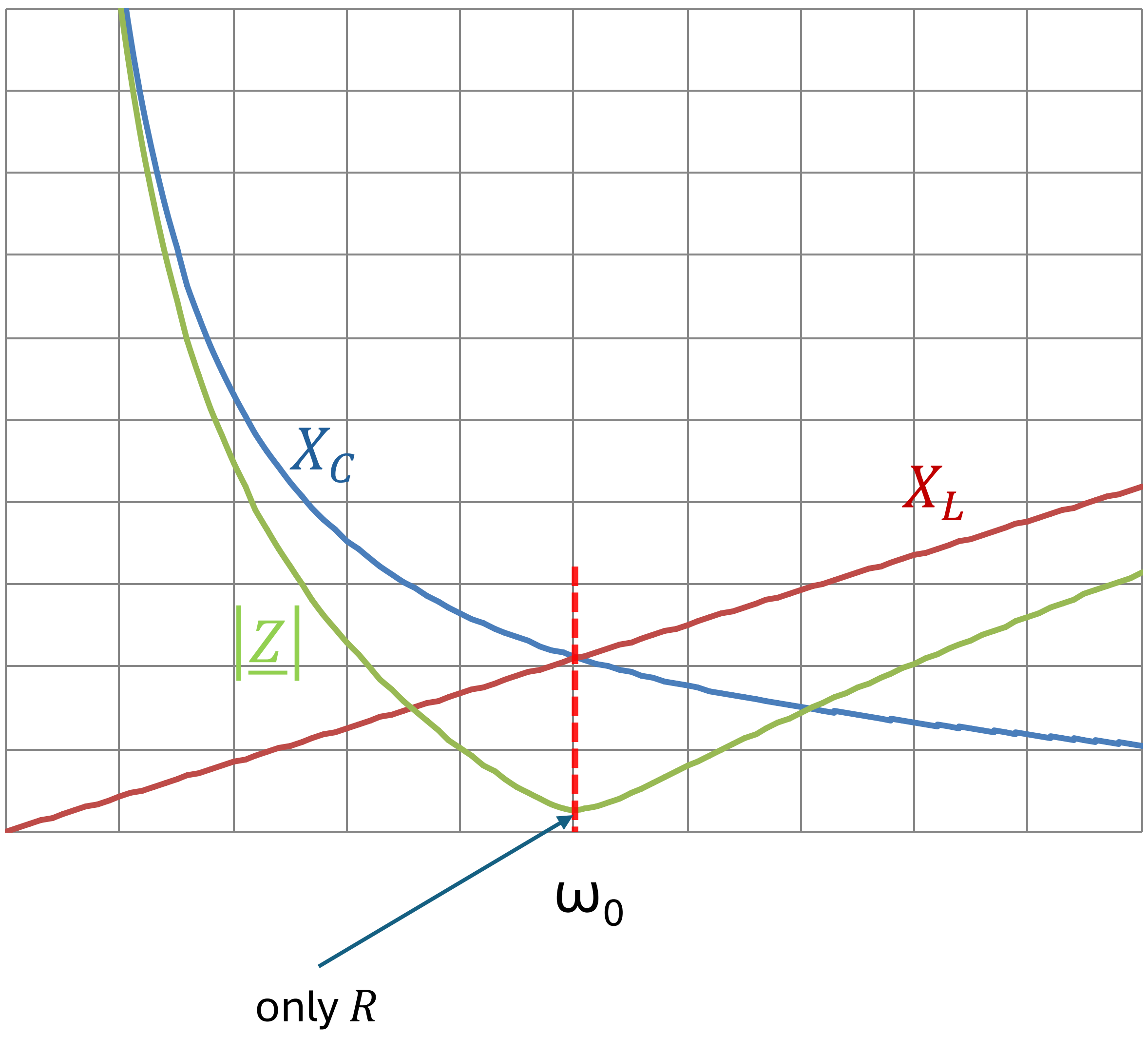
$$\underline{Z}(\omega_0) = R$$

damping

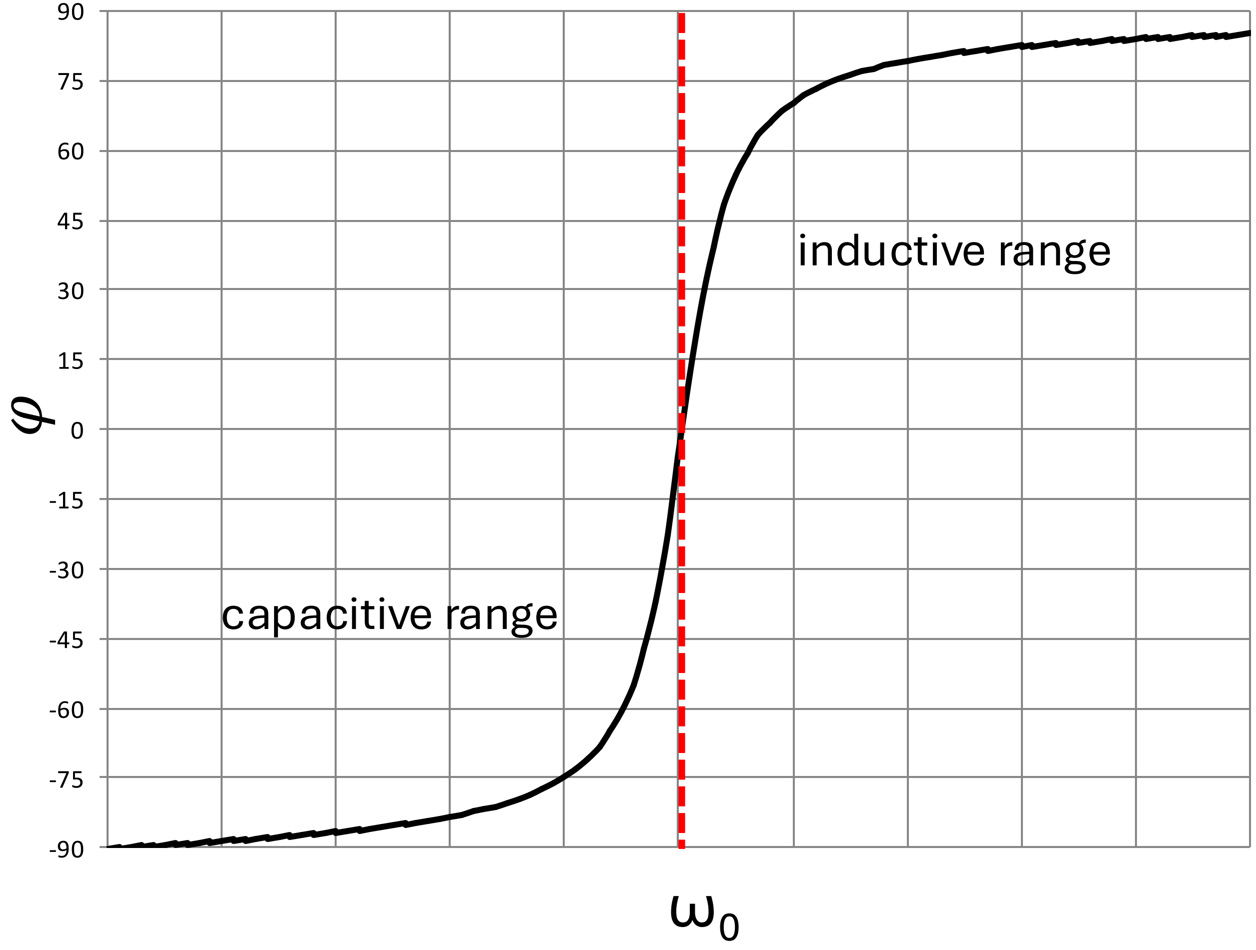


forced oscillation

impedance minimal at resonance:



phase shift between source voltage and current:



Voltage Resonance

$$U_L = U \frac{\omega L}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$U_C = U \frac{1}{\omega C \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

for ω_0: $U_L = U_C = U \frac{1}{R}\sqrt{\frac{L}{C}} = U \cdot Q$

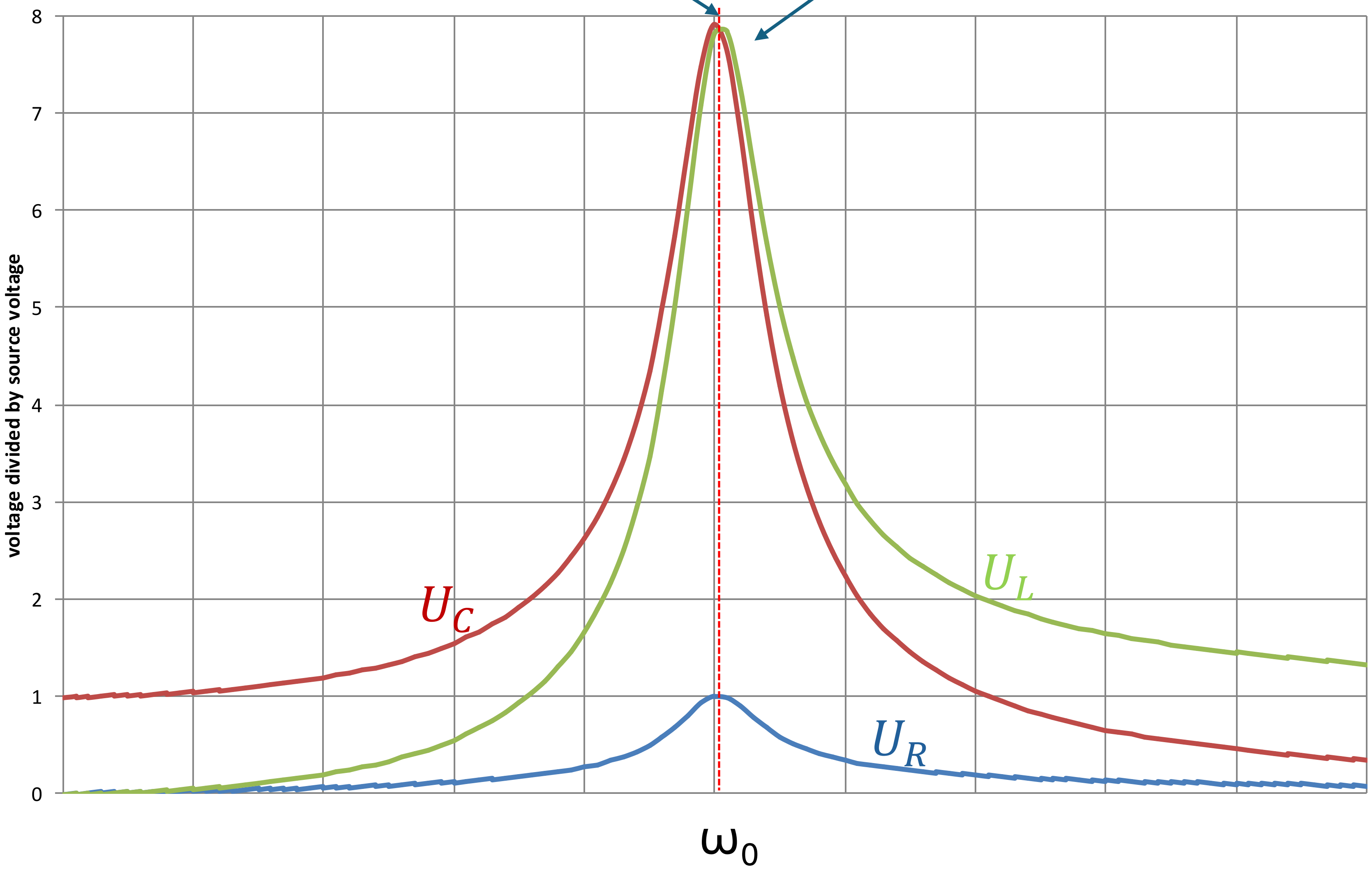
$$\boxed{\underline{U_L} = -\underline{U_C}}$$

quality factor $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$

can be > 1

→ voltages across inductor and capacitor can be significantly larger than the source voltage (resonance catastrophe) phase shift of 180°

maximum of U_L slightly above ω_0
maximum of U_C slightly below ω_0



$U_R = U \frac{R}{\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}}$ → maximum for $\omega=\omega_0$: $U_R = U$ (also maximum current)

Voltage Maxima

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{dU_L}{d\omega_L} = 0 \quad \rightarrow \quad \omega_L = \sqrt{\frac{2}{2LC - R^2C^2}} = \sqrt{\frac{2}{2LC - \frac{LC}{Q^2}}} = \omega_0 \frac{1}{\sqrt{1 - \frac{1}{2Q^2}}}$$

$$\frac{dU_C}{d\omega_C} = 0 \quad \rightarrow \quad \omega_C = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}} = \omega_0\sqrt{1 - \frac{1}{2Q^2}}$$

→ no maxima for $Q \leq \frac{1}{\sqrt{2}}$

resonance behavior more pronounced the higher Q

Parallel Resonant Circuit

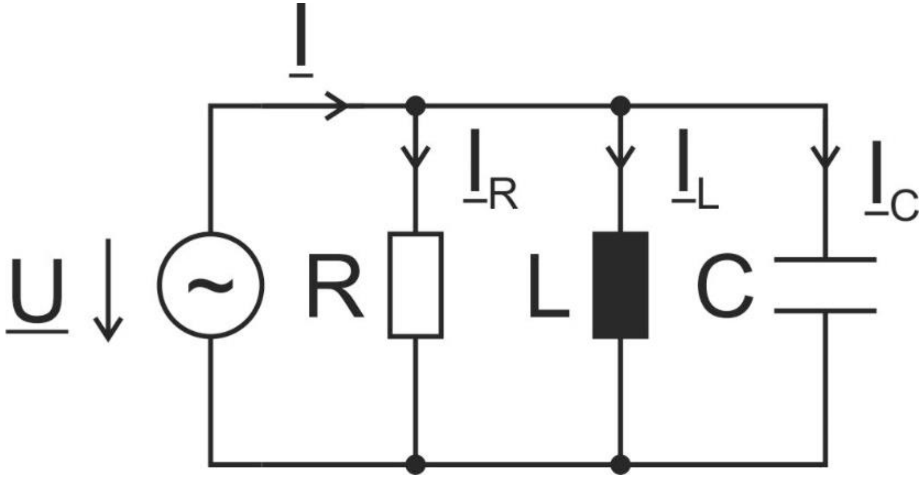
$$\underline{Y} = G + j\left(\omega C - \frac{1}{\omega L}\right)$$

$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$

$$\rightarrow \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\underline{Y}(\omega_0) = G$$

below ω_0: determined by inductance
above ω_0: determined by capacitance



admittance minimal at resonance (impedance maximal) → total current minimal ($\underline{I}_L = -\underline{I}_C$)

Current Resonance

$$I_R = I \frac{G}{\sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$I_L = I \frac{1}{\omega L \sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$I_C = I \frac{\omega C}{\sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

resonance:

$$I_R = I = \frac{U}{R}$$

$$I_L = I_C = IR\sqrt{\frac{C}{L}} = I \cdot Q$$

→ current through inductor and capacitor can be larger than the source current

Current Maxima

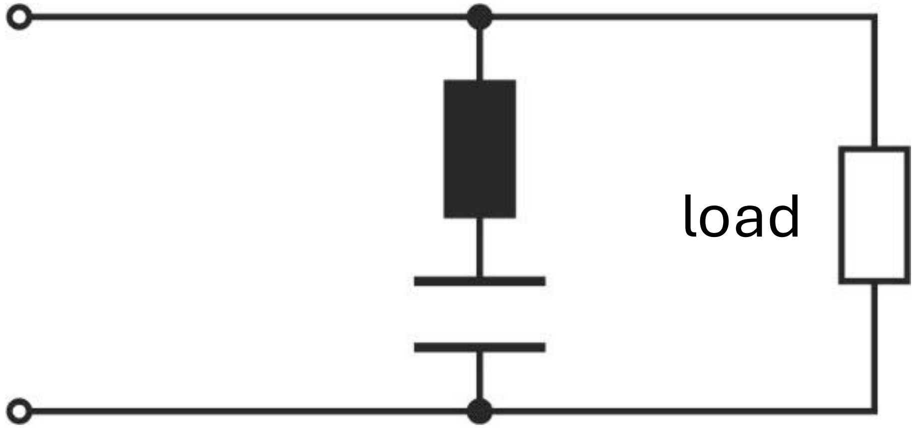
$$\frac{dI_L}{d\omega_L} = 0 \quad \rightarrow \quad \omega_L = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$

$$\frac{dI_C}{d\omega_C} = 0 \quad \rightarrow \quad \omega_C = \omega_0 \frac{1}{\sqrt{1 - \frac{1}{2Q^2}}}$$

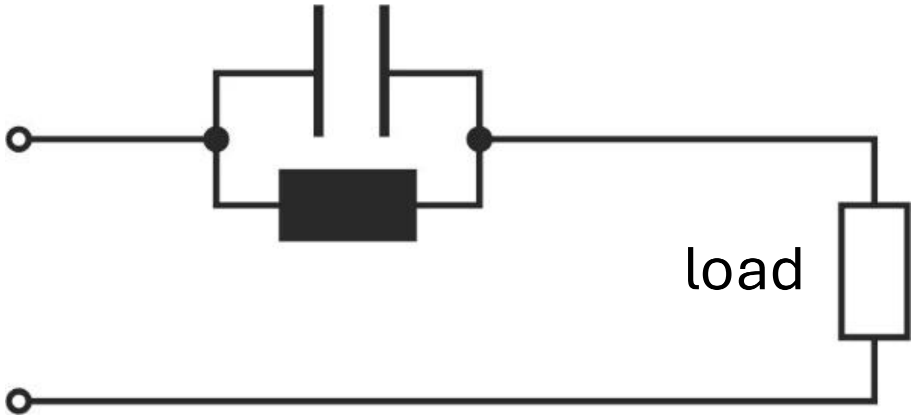
→ again, no maxima for $Q \leq \frac{1}{\sqrt{2}}$

Second-Order Filters

acceptor circuit:
admittance peak at resonance



rejector circuit:
impedance peak at resonance



Three-Phase Circuits

idea:

three alternating currents of the same frequency and amplitude but phase-shifted by 120° from each other

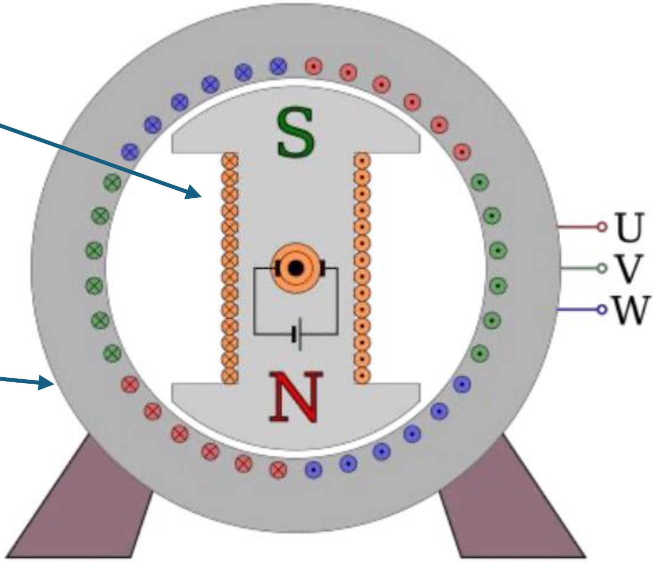
motivation:

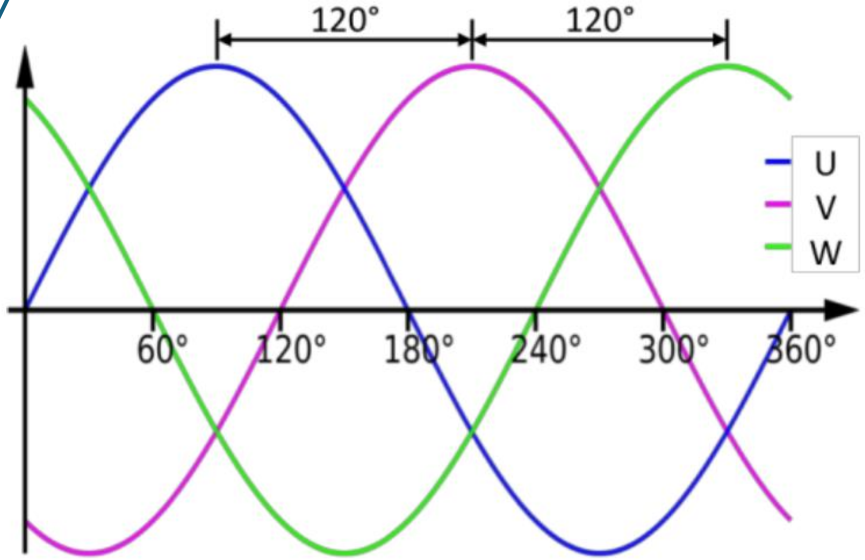
- constant power transmission
- generation of rotating magnetic field → electric motor

Three-Phase Generator

- rotor: (mechanically) rotating magnetic field (generated by coil with DC or permanent magnet)
- stator: three coils offset by 120°
- induces sinusoidal alternating voltage in each coil, phase-shifted by 120°

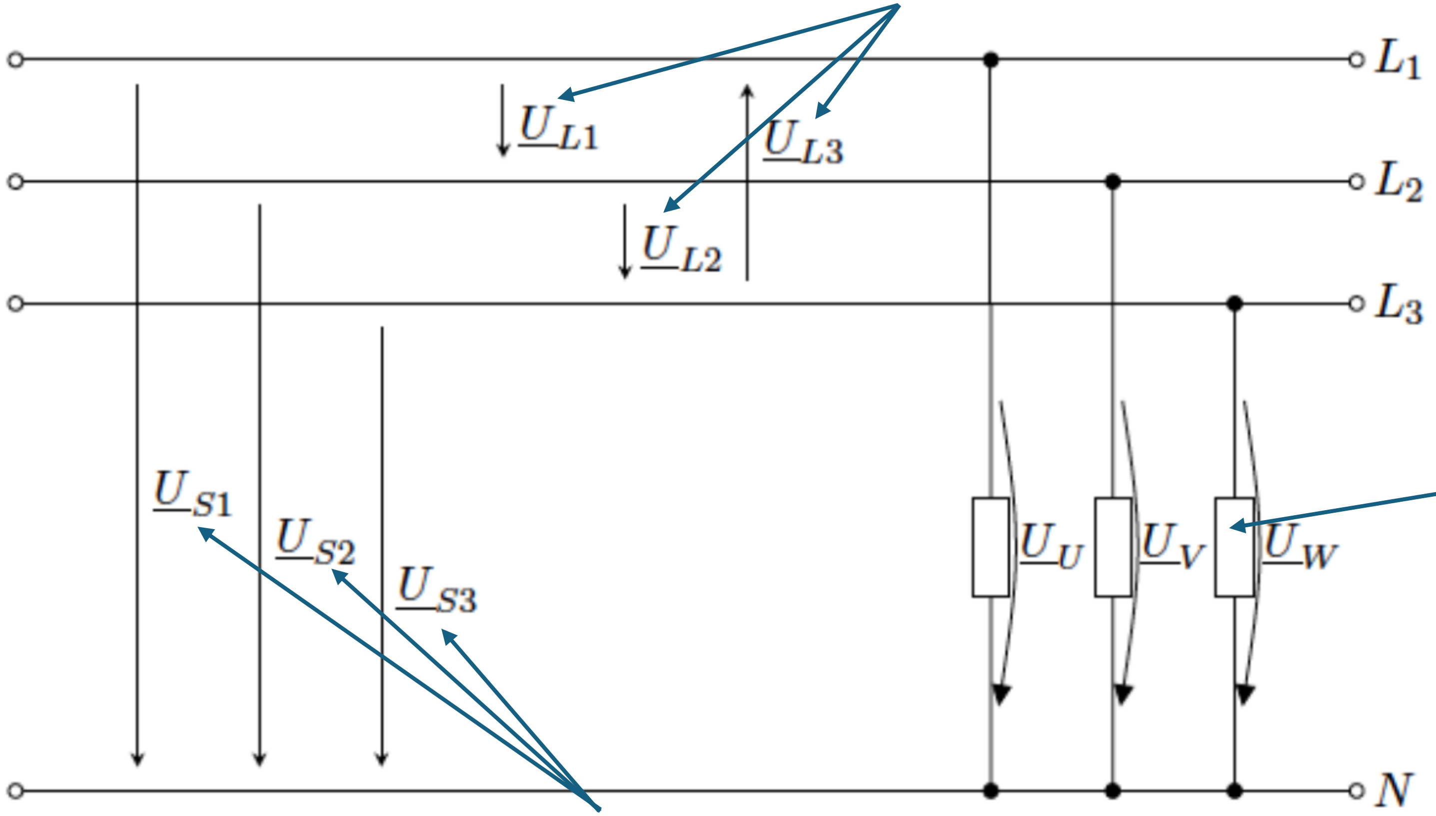
phase identifiers on motor terminals





Three-Phase Network

line voltages (also U_{12}, U_{23}, U_{31})



three phases

loads (e.g., motor)

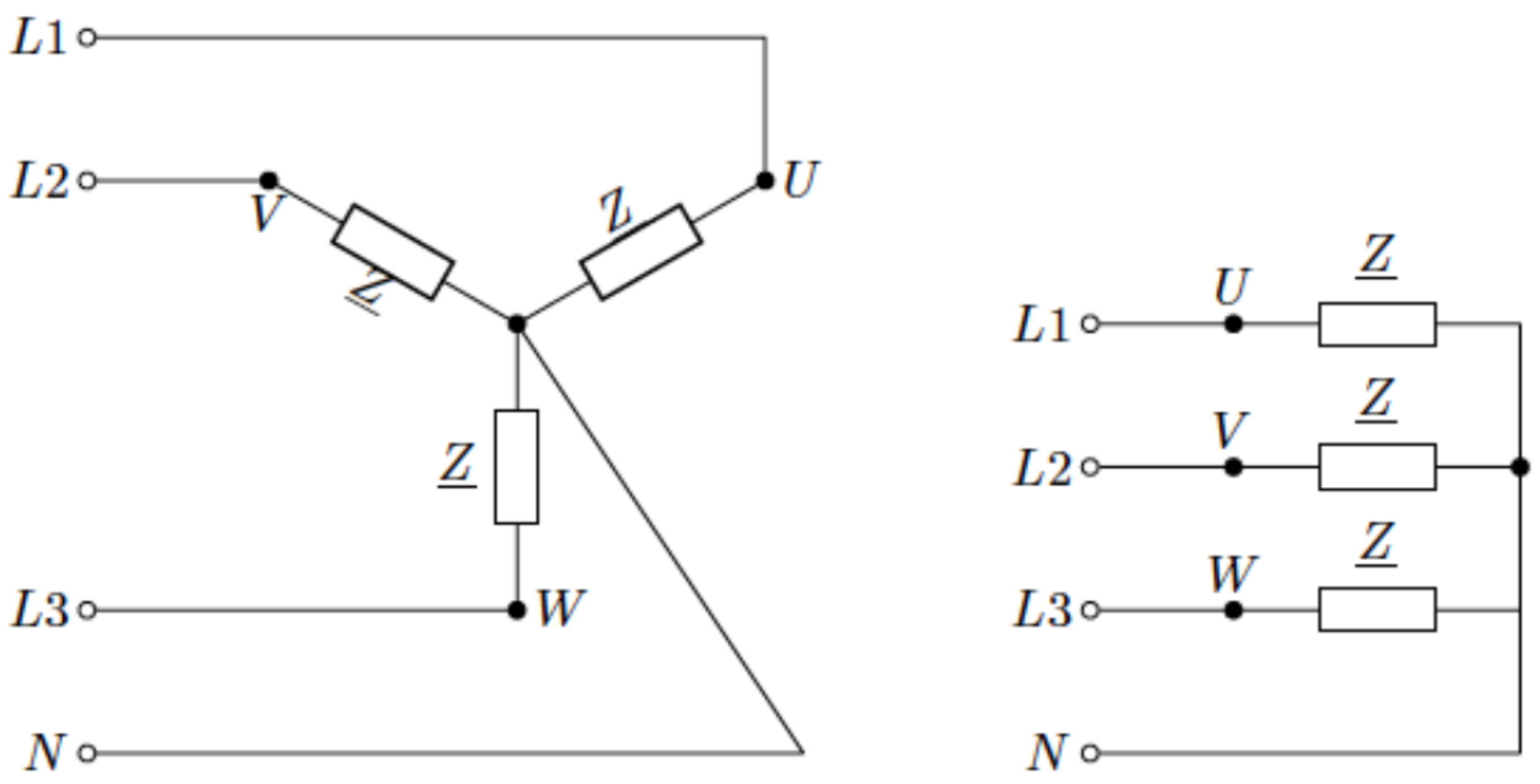
neutral conductor

(optional: return wire not absolutely necessary)

phase voltages

(also U_1, U_2, U_3)

Star (Y) Connection



no current in N for symmetric loads (the three currents add to 0)
→ N could be removed

Phase Voltages

$$u_1 = \hat{u} \cdot \cos(\omega t)$$

$$u_2 = \hat{u} \cdot \cos\left(\omega t - \frac{2}{3}\pi\right)$$

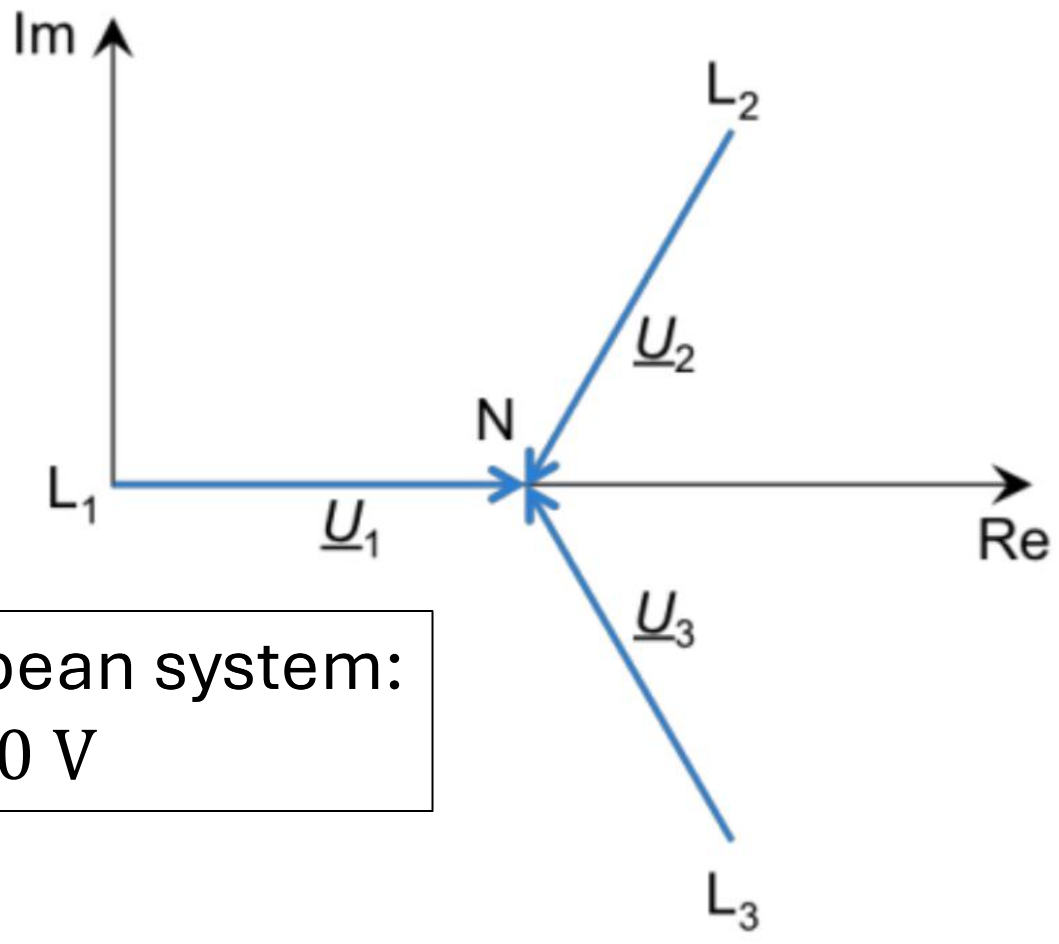
$$u_3 = \hat{u} \cdot \cos\left(\omega t - \frac{4}{3}\pi\right)$$

$$\underline{U_1} = U$$

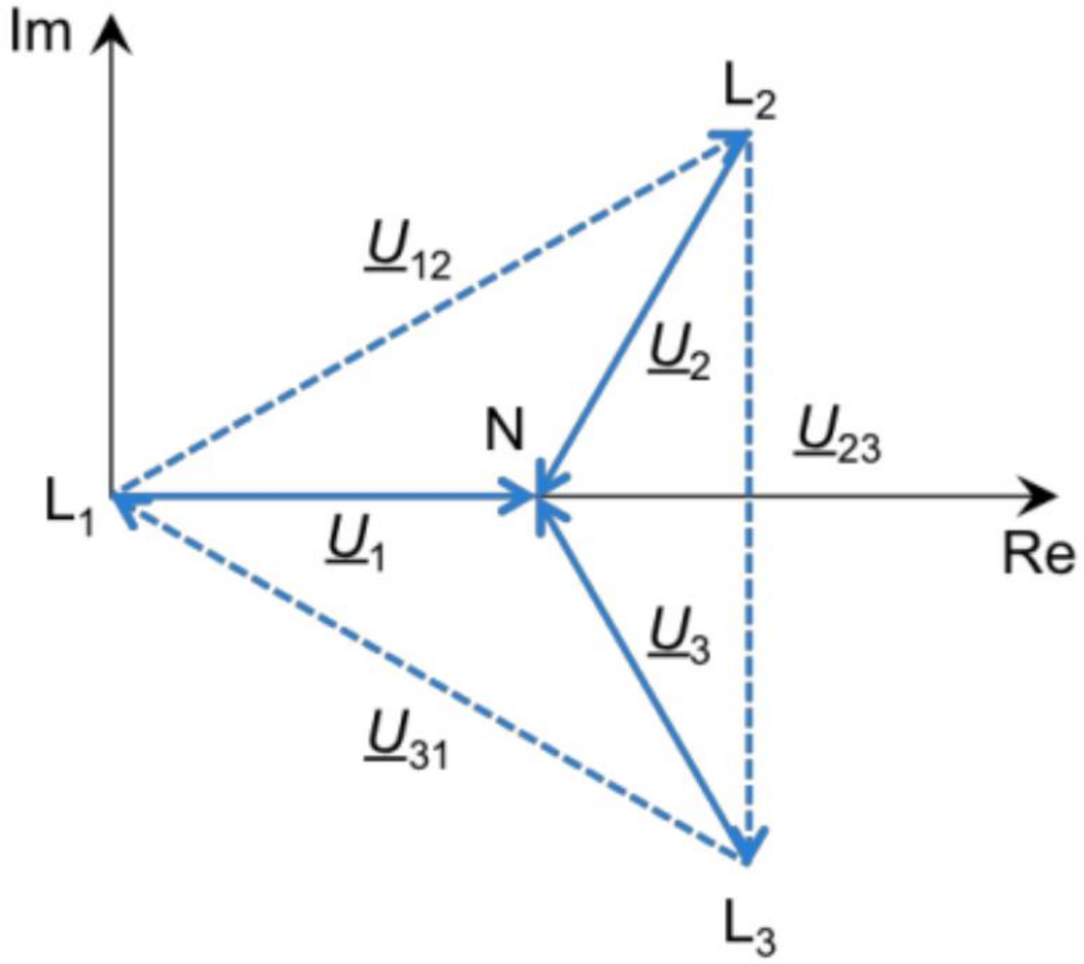
$$\underline{U_2} = U \cdot e^{-j\frac{2}{3}\pi}$$

$$\underline{U_3} = U \cdot e^{-j\frac{4}{3}\pi}$$

typical European system:
230 V



Line Voltages



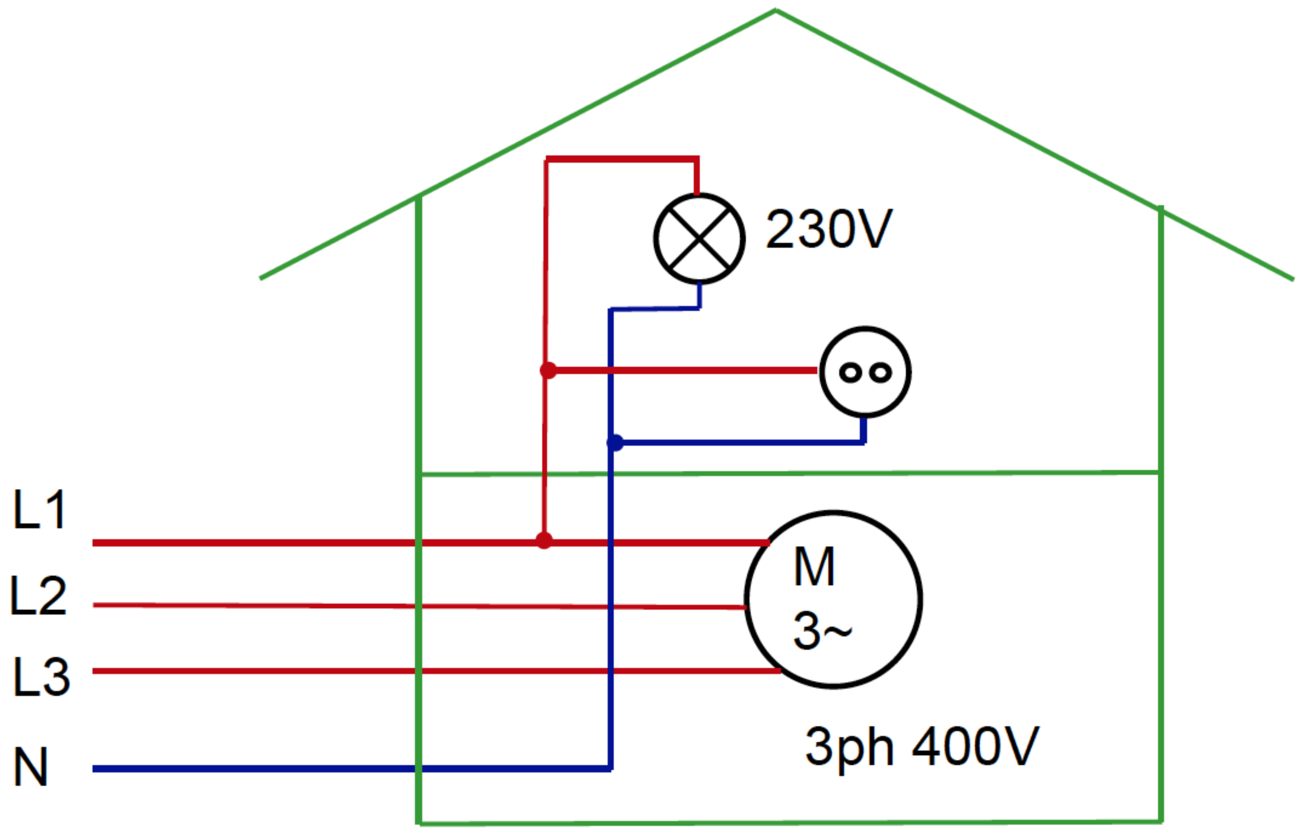
typical European system:
$400\ \mathrm{V}$

$$U_{12} = U_{23} = U_{31} = \sqrt{3} \cdot U$$

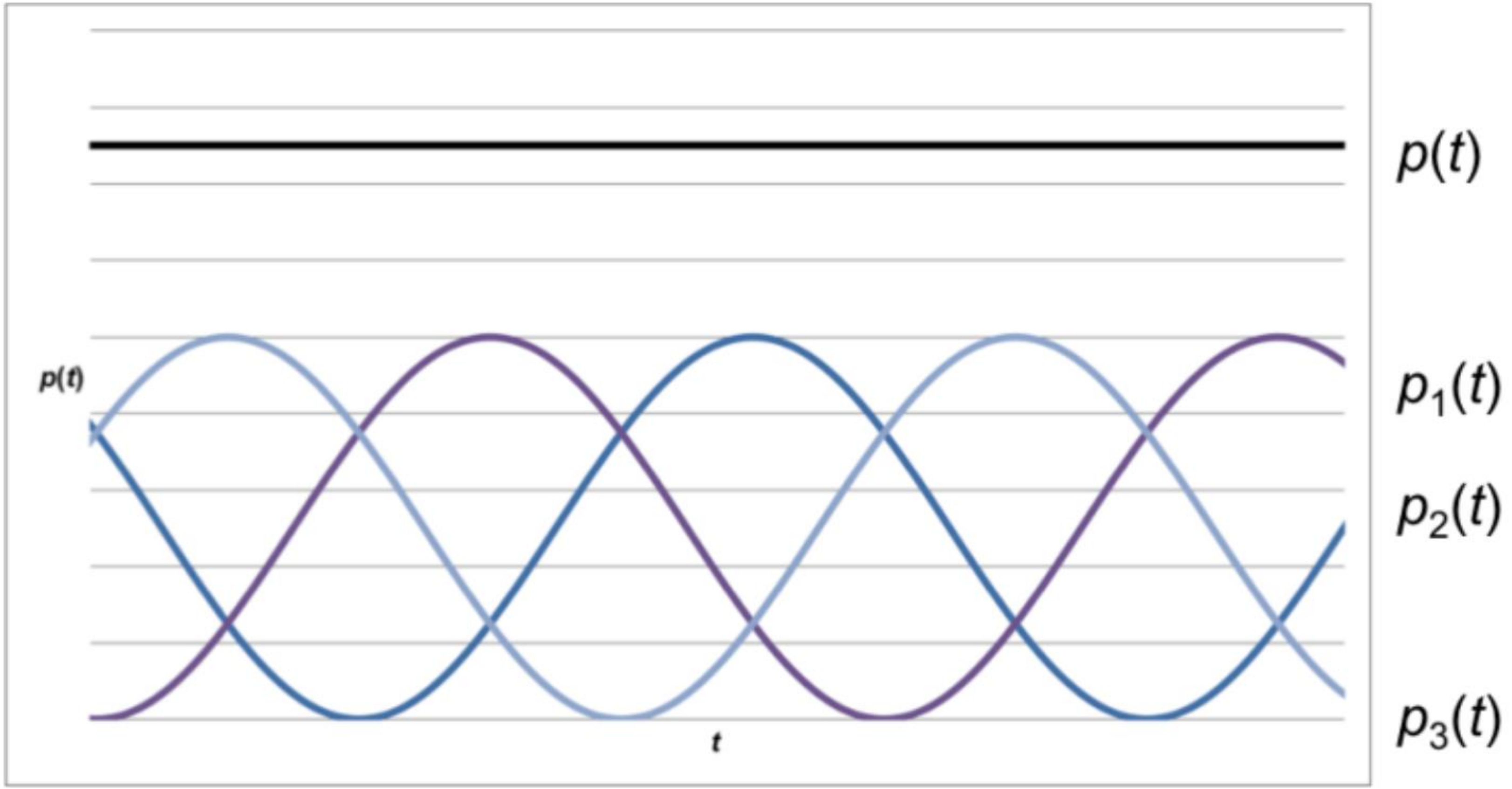
for example:

$$\underline{U_{12}} = \underline{U_1} - \underline{U_2} = U - U \cdot e^{-j\frac{2}{3}\pi} = U \cdot \left(1 - \left(-0.5 - j\frac{\sqrt{3}}{2}\right)\right)$$

$$= U \cdot 1.5 + j\frac{\sqrt{3}}{2} = \sqrt{3} \cdot U \cdot e^{j\frac{\pi}{6}}$$


230V
L1
L2
L3
N
M
3~
3ph 400V

Constant Power

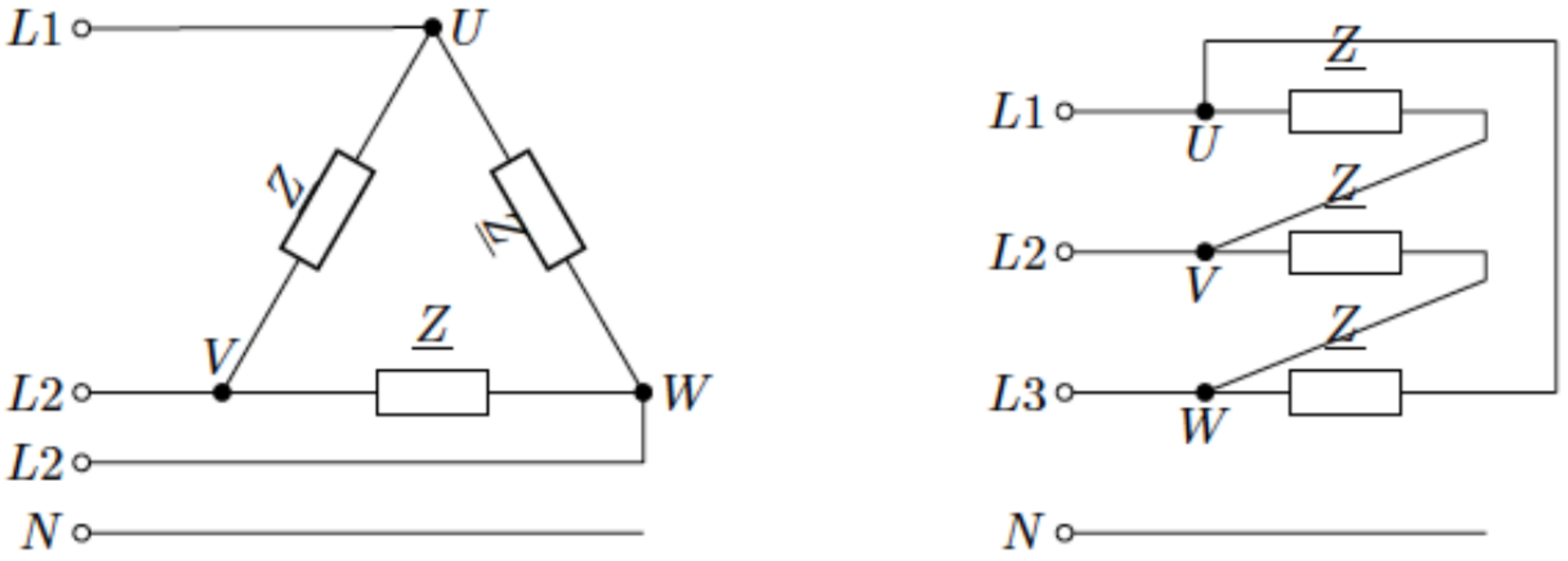


$$p(t) = \frac{u_1^2(t)}{R_1} + \frac{u_2^2(t)}{R_2} + \frac{u_3^2(t)}{R_3}$$

for symmetric load:
$R_1 = R_2 = R_3 = R$

$\rightarrow p(t) = \frac{\hat{U}^2}{R}\left(\cos^2 \omega t + \cos^2\left(\omega t - \frac{2}{3}\pi\right) + \cos^2\left(\omega t + \frac{2}{3}\pi\right)\right) = \frac{3\hat{U}^2}{2R}$

Delta (Δ) Connection



no neutral point

$$U_{12} = U_{23} = U_{31} = U = 400\ \mathrm{V}$$